Robust Mean and Eigenvalues Regularized Covariance Matrix Estimation

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Abstract

We investigate robust mean estimation through utilizing different loss functions, including the Huber, Catoni (Catoni, 2012), and two newly defined loss functions created to gauge the performance of over- or underestimating important "spread" parameters. We allow $E|x_i|^q < \infty$, where $1 < q \leq 2$. This is a new contribution to the literature, when only the case of $q = 2$ has been studied, i.e., the robust mean estimation with the existence of at least second order moments. This development is particularly important when we are estimating higher order moments themselves, like the kurtosis of financial return data. With our proposed robust mean estimator, we create a new eigenvalues-regularized covariance matrix estimator under the assumption $p/n \to c > 0$, where p is the dimension and n is the sample size. This estimator needs only the existence of up to the ξth order moments, $2 < \xi \leq 4$, which is a huge relaxation compared to requiring the existence of 12th order moments in Lam (2016). Theoretical properties are analysed through a newly defined relative loss function, and extensive empirical studies and comparisons with contemporary estimators like the Huber-type M-estimator and rank-based estimator analyzed in Avella-Medina et al. (2018) are performed, showing the superiority of our estimator. A set of ionosphere data is also analyzed in the end.

Key words and phrases. Non-elliptical distribution; Existence of unknown order moments; M-estimator; Robust covariance nonlinear shrinkage; Non-patterned covariance matrix estimation.

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1 Introduction

With the volume and variety of data ever increasing in this era of "big data", assuming the existence of higher order moments can be dangerously wrong in applications such as financial return analysis, or data sets with unusually large number of "outliers". The latter can mean a systematic change to the nature of the data generating process than one would expect, such that the distribution of the data can have much heavier tail than sub-Gaussian or sub-exponential counterparts. Traditional statistics like the sample mean and sample covariance matrix can suffer huge drop in performance as a result.

In robust statistics, a very useful idea is to model the data as partly "regular" and partly "outliers". A recent example is Chen et al. (2018). They consider such a model with the proportion of outliers ϵ goes to 0 while the regular distribution is normal (termed an ϵ-contamination model) in order to derive statistical properties of their estimators. The seminal paper Huber (1964) is one of the first to consider such a model for location parameter estimation. A loss function approach is also proposed for finding M-estimators, with the famous Huber loss (Huber, 1972) derived from considering the ϵ-contamination model in a "least favorable" sense. While certain minimax properties are proved, the Huber loss is optimal for a "contaminated" normal distribution. When the data distribution is wildly different from normal, or even asymmetric, there is no guarantee that the Huber loss leads to good performance of the resulting M-estimator.

Considering only the properties of an influence function (i.e., the derivative of a loss function), Catoni (2012) derives a non-asymptotic probability bound for the resulting M-estimator, enabling the construction of confidence interval. This is a breakthrough in not only that the result is non-asymptotic, but also does not need a "contamination" model for the data. It means that the data distribution can be far from normal, or even asymmetric. Although the same technique is used in Fan et al. (2017) to prove a similar non-asymptotic bound for mean and variance M-estimators for the Huber loss, both papers assume the existence of second order moments for a location M-estimator, while the explicit comparisons between the two is lacking both theoretically and empirically.

As one of the major contribution in this paper, we create new classes of loss functions to allow for a similar non-asymptotic bound as in Catoni (2012), but allow the data to have only moments up to the q_0th order, where $1 < q_0 \leq 2$. Since q_0 and the true "spread" of the data v_0 are not known, we also investigate the effect of over/under-specifying these parameters as another major contribution. The conclusion is that we want to overestimate q_0 and v_0 as opposed to underestimating them for reducing the length of the confidence interval for the resulting M-estimator. In the most likely situation where we assumed that second order moments exist but in fact only up to the q_0th order moments do, the spread parameter v_0 should be overestimated by a factor which is diverging to infinity as the sample size in order to bring the rate of convergence of the M-estimator closer to optimum. While Avella-Medina et al. (2018) also propose a Huber-loss M-estimator for the mean requiring only up to $(1+\epsilon)$th order moments to exist, they have not provided an estimator of ϵ for implementing their approach since their optimal tuning parameter H depends on ϵ. Our mean M-estimator using the newly proposed class of influence function nicely complements this, since we always assume the second order moments to exist even if in fact $q_0 < 2$, and we fix the rate of divergence of the tuning parameter to be practically at $n^{1/3}$ (see Section 2.4), with theoretical results showing the effect of using other powers of n as well (see Theorem 3). For the details of our proposed M-estimator, please see Section 2 and the results therein.

Another major contribution of the paper comes from applying our proposed M-estimator for nonlinear shrinkage of large covariance matrix when the data only have up to the ξth order moments, $2 < \xi \leq 4$. Nonlinear shrinkage is first introduced in Ledoit and Wolf (2012), where for a class of rotation-equivariant estimator, we find the one with eigenvalues nonlinearly shrunk so as to minimize a certain loss function. In Lam (2016), with the assumption of having at least 12th order moments for the data, it is proved that their estimator enjoys almost sure positive semi-definiteness as sample size $n \to \infty$ under the setting $p/n \to c > 0$, where p is the dimension of a data vector. The data splitting method employed in Lam (2016), originally proposed in Abadir et al. (2014), in fact enables us to extend their nonlinear shrinkage estimator to a robust one in this paper.

Focusing on sparse covariance matrix, Avella-Medina et al. (2018) also proposed a robust covariance matrix estimator with only $(2+\epsilon)$th order of moments for the data. However, their covariance matrix estimator is not necessarily positive semi-definite, and the estimation of ϵ, like its mean counterpart, is not discussed but is unfortunately needed in practice for setting a tuning parameter. Our nonlinear shrunk estimator has the appeal of being positive semi-definite by construction, and is almost surely positive definite asymptotically under $p/n \to c > 0$ (see Theorem 4), while we fix the overestimation factor of the tuning parameter at $n^{1/3}$ like our mean counterpart. It is already enough to show that our estimator is at least almost surely as good as a nonlinear shrinkage counterpart asymptotically. In practice, our estimator outperforms the nonlinear shrinkage estimator when the data distribution (symmetric or asymmetric) has a very fat tail, and also outperform Huber-loss type estimator in Avella-Medina et al. (2018). See Section 5 for more details.

The rest of the paper is organized as follows. Section 2 introduces the generalized Catoni loss for mean estimation, and the non-asymptotic probabilistic bound on the resulting M-estimator. Effects of estimating q_0 and the spread parameter is illustrated through another non-asymptotic probabilistic bound. The insights lead us to create two new families of influence functions and some practicalities laid down for improving our M-estimator for the mean. Section 3 presents the need for regularization when dimension is large, and propose a robust version of nonlinear shrinkage for large covariance matrix estimation. Section 4 presents the corresponding theoretical results and practical implementations. Extensive simulation results for comparing our M-estimators to other state-of-the-art robust estimators are presented in Section 5. A set of ionosphere data is analyzed in the same section as well, using our proposed estimators for classification purpose. All proofs are given in Section 6.

2 Robust mean estimation

Suppose we have a random sample $\{x_i\}_{i=1,\dots,n}$, with $E(x_i) = \mu$ and

$$v := E^{1/q}|x_i - \mu|^q, \quad 1 < q \leq 2, \tag{2.1}$$

where $E|x_i|^k$ for any $k > q$ do not necessarily exist (we still use $q = 2$ when Ex_i^k exists for $k > 2$). In Catoni (2012) and Fan et al. (2017), they use $q = 2$, meaning that second order moments exist by assumption. There are however a lot of practical problems that we do not want to assume the existence of moments much beyond the first order. For instance, it is well-known in finance that moments much beyond the fourth order for asset returns do not exist, especially over a volatile period. If we want to estimate the skewness or even the kurtosis of an asset return $\{r_i\}$, assuming the existence of second order moments of the raw data ($\{r_i^3\}$ and $\{r_i^4\}$ respectively) means assuming the existence of the 6th or 8th order moments respectively for $\{r_i\}$, which can be problematic.

2.1 Generalized Catoni loss

Consider a non-decreasing influence function $\psi : \mathbb{R} \to \mathbb{R}$ such that

$$-\log(1 - x + c|x|^q) \leq \psi(x) \leq \log(1 + x + c|x|^q). \tag{2.2}$$

It is easy to see that the smallest c allowed is $c = c_q := q^{-q/2}(2-q)^{1-q/2}(q-1)^{q-1}$. We show in Lemma 3 in Section 6 that

$$\psi_q(x) := \mathrm{sgn}(x)\big\{ -\log(1 - |x| + c_q|x|^q)\mathbf{1}\{|x| < b_q\} + a_q\mathbf{1}\{|x| \geq b_q\}\big\}, \tag{2.3}$$

where $b_q := (qc_q)^{1/(1-q)}$ and $a_q := -\log(1 - b_q + c_q b_q^q)$, is one of the narrowest possible influence function satisfying (2.2). The function $\mathbf{1}\{\cdot\}$ is the indicator function, and $\mathrm{sgn}(x)$ returns the sign of x, which is 0 when $x = 0$. Comparing ψ_q with equation (2.3) in Catoni (2012), we call ψ_q the influence function of the generalized Catoni loss, which is defined as

$$\ell_q(x) = \int_0^x \psi_q(s)ds. \tag{2.4}$$

The motivation for the bounds in (2.2) has two folds. Firstly, Fan et al. (2017) and Catoni (2012) both used $q = 2$ in their bounds which lead to allowing $Ex_i^2 < \infty$ for the data. Considering $q < 2$ allows for the weaker assumption $E|x_i|^q < \infty$ only, which is particularly important for data sets without second order moments, or when we need to estimate higher order moments for the data. Secondly, the value c gives us a potential tuning parameter for better performance. It turns out this allows us to gauge the performance of the estimators when we under/overestiamte the value of q, or another needed parameter α. See Theorem 2 and 3.

2.2 Robust mean estimator and probabilistic bounds

With the generalized Catoni loss defined in (2.4), we define the robust mean estimator $\widehat{\mu}_\alpha$ of μ as the solution to

$$\sum_{i=1}^n \psi_q(\alpha(x_i - \theta)) = 0,$$

where $\alpha > 0$ is a tuning parameter to be defined more precisely in Theorem 1, and ψ_q is the influence function defined in (2.3). Hence $\widehat{\mu}_\alpha$ is the minimizer of

$$L_q(\theta) := \sum_{i=1}^n \ell_q(\alpha(x_i - \theta)), \tag{2.5}$$

which has a unique minimum since ℓ_q is convex as a result of ψ_q being non-decreasing.

Theorem 1. *Let $E|x_i|^q < \infty$ for a random sample $\{x_i\}_{i=1,\dots,n}$ with mean μ, $1 < q \le 2$. Assume that*

$$\frac{\log(1/\delta)}{n} \le \frac{q-1}{q}\left(\frac{k-1}{k^q d_q c_q}\right)^{1/(q-1)}, \text{ where } d_q = 2^{q-1}.$$

and $1 < k \le 2$ is a constant. If we let

$$\alpha = \left(\frac{1}{d_q c_q (q-1)}\right)^{1/q} \frac{1}{v} \left(\frac{\log(1/\delta)}{n}\right)^{1/q},$$

where $v \geq E^{1/q}|x_1 - \mu|^q$*, then we have*

$$P\left(|\widehat{\mu}_\alpha - \mu| \geq \frac{kqd_q^{1/q}c_q^{1/q}v}{(q-1)^{1-1/q}}\left(\frac{\log(1/\delta)}{n}\right)^{1-1/q}\right) \leq 2\delta.$$

The proof of the Theorem is relegated to Section 6. Several observations can be made regarding Theorem 1. Firstly, when q gets closer to 1, α will be larger in general. In view of Figure 1 for the influence function $\psi_q(\alpha(x-\theta))$ and the loss function $\ell_q(\alpha(x-\theta))$, a larger α "curbs" the contribution to the loss L_q in (2.5) from more "extreme" data points that deviate too much from $\widehat{\mu}_\alpha$ (by restricting a "large" data point x to contribute to the loss only linearly). This certainly makes sense, since a smaller q means that the data distribution has fatter tails, and so there are expected to be more "extreme" data points. Secondly, the rate $(\log(1/\delta)/n)^{1-1/q}$ is slower as q is closer to 1, reflecting the difficulty in estimating μ with limited existence of moments. Thirdly, observe that overestimating $v_0 = E^{1/q}|x_1 - \mu|^q$ is fine, since as long as $v \geq v_0$, the same probability bound holds, just that the error bound will then be larger than it could have been. See Theorem 3 in Section 2.3 for the effects of over- or underestimating v_0.

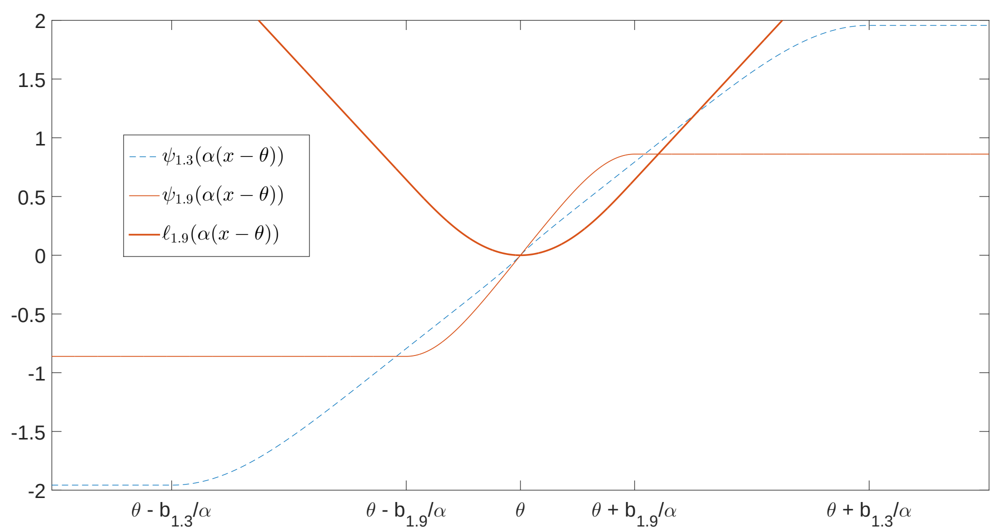

Figure 1: The influence functions $\psi_{1.3}(\alpha(x-\theta))$ and $\psi_{1.9}(\alpha(x-\theta))$, and the loss function $\ell_{1.9}(\alpha(x-\theta))$. Beyond $\theta \pm b_{1.9}/\alpha$, $\ell_{1.9}(\alpha(x-\theta))$ is linear in x.

Also, the error bound in the probabilistic statement of $|\widehat{\mu}_\alpha - \mu|$ is in general shorter than using Huber loss (see Fan et al. (2017) for instance). When $q = 2$, the bound is

$$\sqrt{2}kv\sqrt{\frac{\log(1/\delta)}{n}},$$

which is smaller than $4v(\log(1/\delta)/n)^{1/2}$ in Theorem 5 of Fan et al. (2017). In fact, this bound is exactly comparable to η in Proposition 2.4 of Catoni (2012) when $q = 2$. Hence Theorem 1 can be considered a generalization to Proposition 2.4 of Catoni (2012) when $1 < q < 2$. Finally, with fixed δ, n and known v and q, we can solve for k using the equality

$$\frac{\log(1/\delta)}{n} = \frac{q-1}{q}\left(\frac{k-1}{k^q d_q c_q}\right)^{1/(q-1)}$$

to know the exact length of the shortest confidence interval for $\widehat{\mu}_\alpha$ at confidence level $1 - 2\delta$.

Remark 1. Theorem 1 has a parallel version for the Huber loss. Consider the influence function $\psi(x)$ defined by $\psi(x) = x$ for $|x| < 1$, $\psi(x) = -1, x \leq -1$ and $\psi(x) = 1, x \geq 1$. The Huber loss is defined as $\ell_h(x) = \int_0^x \psi(x)dx$. Let $\widehat{\mu}_{\alpha,h}$ be the minimizer of $L_h(\theta) := \sum_{i=1}^n \ell_h(\alpha(x_i - \theta))$.

Lemma 1. *Define* $c_{q,h} = \max(e-2, q^{-q}(q-1)^{q-1}(1-e^{-1})^{1-q})$*,* $1 < q \leq 2$*. Assume* $E|x_i|^q < \infty$ *for a random sample* $\{x_i\}_{i=1,\ldots,n}$ *with mean* μ*. Then Theorem 1 holds true with* c_q *and* $\widehat{\mu}_\alpha$ *there replaced by* $c_{q,h}$ *and* $\widehat{\mu}_{\alpha,h}$ *respectively.*

We always have $c_{q,h} > c_q$ for $1 < q \leq 2$, and hence the confidence interval using the Huber loss is longer than the generalized Catoni loss in general. For $q = 2$ especially, the confidence interval (with confidence level $1 - 2\delta$) for μ indicated by Lemma 1 has length $4k(e-2)^{1/2}v(\log(1/\delta)/n)^{1/2}$. This is longer than the corresponding one for $\widehat{\mu}_\alpha$ under Theorem 1 when $q = 2$.

2.3 Effects of estimating q and v

In practice, we may not know the true value of q and v, say q_0 and $v_0 = E^{1/q_0}|x_1 - \mu|^{q_0}$ respectively. To gauge the performance of robust mean estimation when q_0 and v_0 are estimated, we first show the following.

Theorem 2. *For all* $1 < q \leq 2$ *and* $x \in \mathbb{R}$*, we have*

$$-\log\Big(1 - x + \frac{5}{4}c_q|x|^q\Big) \leq \psi_2(x) \leq \log\Big(1 + x + \frac{5}{4}c_q|x|^q\Big).$$

Assume $E|x_i|^{q_0} < \infty$ *and* $v_0 = E^{1/q_0}|x_1 - \mu|^{q_0}$*, where* $\{x_i\}_{i=1,\ldots,n}$ *is a random sample with mean* μ*. Let* $\widehat{\mu}_\alpha$ *be the minimizer of* $L_2(\theta)$ *defined in (2.5). Then for* $v \geq v_0$*, Theorem 1 holds true with* q *and* c_q *there replaced by* q_0 *and* $5c_{q_0}/4$ *respectively.*

Proof of Theorem 2 will be given in Section 6. See Figure 2 for an illustration of the inequalities in the theorem. Theorem 2 shows that even when $q_0 < 2$, we can still use the influence function ψ_2 from the Catoni loss for robust mean estimation, not necessarily ψ_{q_0}. Of course, the process of robust mean estimation requires that we specify the value of α, which still needs estimators of q_0 and v_0. The result of Theorem 2 regarding $\widehat{\mu}_\alpha$ assumed the knowledge of q_0 and v_0.

To gain more insights into the effects of estimating q_0 and v_0, let q be a constant replacing q_0 such that

$$\frac{1}{q} = \frac{1}{q_0} - \eta,$$

where η can be positive (overestimation) or negative (underestimation). Let u be a constant replacing v_0.

Theorem 3. *Assume the settings in Theorem 2, but we now use*

$$\alpha' = \left(\frac{4}{5d_q c_q(q-1)}\right)^{1/q} \frac{1}{u} \left(\frac{\log(1/\delta)}{n}\right)^{1/q}$$

as a tuning parameter in minimising $L_2(\theta)$ in (2.5), resulting in $\widehat{\mu}_{\alpha'}$. Define

$$\begin{aligned}\beta_{n,q,u} &:= \left(\frac{5(q-1)d_q c_q}{4}\right)^{1/q} \left(\frac{\log(1/\delta)}{n}\right)^{1-1/q_0} \\ &\cdot \left\{\frac{5}{4} c_{q_0} d_{q_0} v_0 \left(\frac{v_0}{u}\right)^{q_0-1} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \left(\frac{\log(1/\delta)}{n}\right)^{-\eta(q_0-1)} + u\left(\frac{\log(1/\delta)}{n}\right)^{\eta}\right\}.\end{aligned}$$

Then for a constant $1 < k \leq 2$, if

$$\frac{\log(1/\delta)}{n} < \left(\frac{4(k-1)}{5k^{q_0} c_{q_0} d_{q_0}}\right)^{q/(q_0(q_0-1))} \left(1 + \frac{5}{4} c_{q_0} d_{q_0} \left(\frac{v_0}{u}\right)^{q_0} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q}\right)^{-q/q_0},$$

we have $P\big(|\widehat{\mu}_{\alpha'} - \mu| \geq k\beta_{n,q,u}\big) \leq 2\delta$.

Proof of this theorem can be found in Section 6. Some important observations can be made from this Theorem. When $\eta = 0$, the rate in $\beta_{n,q,u}$ is $(\log(1/\delta)/n)^{1-1/q_0}$, which is the best rate achievable as in Theorem 1 when $q = q_0$. Recall that $1 < q_0 \leq 2$, so that $0 < q_0 - 1 \leq 1$. Hence for large n, we have

$$\left(\frac{\log(1/\delta)}{n}\right)^{-|\eta(q_0-1)|} < \left(\frac{\log(1/\delta)}{n}\right)^{-|\eta|},$$

where the left hand side above is the dominant rate inside the curly bracket in $\beta_{n,q,u}$ when $\eta > 0$, and the right hand side is the corresponding dominant rate when $\eta < 0$. Hence, given the same magnitude $|\eta|$, to make $\beta_{n,q,u}$ smaller, we would want a smaller dominant rate by making $\eta > 0$. In other words, between choosing overestimating q_0 slightly and underestimating q_0 slightly, we would want to choose overestimation for a better rate in $\beta_{n,q,u}$.

Given q_0 is slightly overestimated by q ($\eta > 0$), we would want to overestimate v_0 as well when n is large. In fact, if the overestimation is to the extent that

$$u = bv_0 \left(\frac{\log(1/\delta)}{n}\right)^{-\eta} \tag{2.6}$$

for some constant $b > 0$, we then have

$$\beta_{n,q,u} := \left(\frac{5(q-1)d_q c_q}{4}\right)^{1/q} v_0 \left(\frac{\log(1/\delta)}{n}\right)^{1-1/q_0} \left\{\frac{5c_{q_0} d_{q_0}}{4b^{q_0-1}} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} + b\right\},$$

so that the optimal rate for $\beta_{n,q,u}$ is recovered. When $\eta < 0$, the reverse happens, and we would want to underestimate v_0. When $\eta = 0$, over- or underestimating u are equally "bad" in terms of the rate for $\beta_{n,q,u}$. In practice, we still want to overestimate v_0 slightly when $\eta = 0$, since this makes α smaller, which in turn allows for more efficient use of the whole data set by not "curbing" the effects of as many data points compared to when α is large. With a known $q_0 = 2$, Fan et al. (2019) used the same scheme in the context of covariance matrix estimation (note their α corresponds to our $1/\alpha$). As a special case, when $\eta = 0$, the rate $(\log(1/\delta)/n)^{1-1/q_0}$ in $\beta_{n,q,u}$ actually coincides with that in Proposition 6 and Corollary 1 of Avella-Medina et al. (2018).

2.4 Practical Implementation

With the insights from Theorem 2, we use the Catoni loss for our robust mean estimation. For setting the value of α as α' in Theorem 3, if we set $q = 2$ regardless of the true value q_0, η in (2.6) will be at most very close to $1/2$ since $1 < q_0 \leq 2$. It means that we would at most want to overestimate v by an order of $n^{1/2}$. When $q_0 = 2$, we have $\eta = 0$, but the effects of overestimating v_0 is much diluted anyway when $Ex_i^2 < \infty$, since the number of "outliers" is much reduced compared to when q_0 is close to 1. With these observations, we set $q = 2$ in α, and use

$$\begin{aligned} v &= (|\kappa_{t_0}| + 1)(an^{1/3} + b) \times [100n^{-1/2}\%\text{-trimmed mean of } \{(x_i - t_0)^2\}_{i=1,\ldots,n}]^{1/2}, \text{ where} \qquad (2.7)\\ t_0 &= 10\%\text{-trimmed mean of } \{x_i\}_{i=1,\ldots,n}, \\ \kappa_{t_0} &= \frac{50n^{-1/2}\%\text{-trimmed mean of } \{(x_i - t_0)^3\}_{i=1,\ldots,n}}{[50n^{-1/2}\%\text{-trimmed mean of } \{(x_i - t_0)^2\}_{i=1,\ldots,n}]^{3/2}}, \end{aligned}$$

and a and b are constants to be determined using, e.g., cross-validation. We call $an^{1/3} + b$ the multiplying factor for v. Depending on applications, a and b can be pre-determined as well. Our empirical examples in Section 5 have $a = 0.1, b = 1$ for robust mean estimation and $a = 2, b = -3$ for robust shrinkage covariance estimation. In general, for mean estimation, our experience shows that a should be set to be much smaller than when in robust covariance matrix estimation.

The above proposal works on the fact that the square root of the trimmed mean of $\{(x_i - t_0)^2\}_{i=1,\dots,n}$ will be smaller than the true value v_0 in general, because of the trimming of the extreme observations. But it also provides a much stabler estimator than just taking the usual variance under the presence of extreme observations. The value of κ_{t_0} reflects the skewness of the distribution, since the usual skewness does not exist for $q_0 \leq 2$. The more skewed the distribution, the more severe the underestimation of v is. Hence an adjustment factor incorporating the skewness seems sensible, and κ_{t_0} does the job well.

We also compare the Catoni loss to the performance of two other influence functions. For $\lambda > 0$, define

$$\varphi_{1,\lambda}(x) = \mathrm{sgn}(x)\Big(\log\big(-\lambda + |x| + (1+\lambda)\sqrt{1+x^2}\big)\mathbf{1}\{|x| < b_\lambda\} - \log(\sqrt{\lambda^2+2\lambda} - \lambda)\mathbf{1}\{|x| \geq b_\lambda\}\Big), \quad (2.8)$$

$$\varphi_{2,\lambda}(x) = -\mathrm{sgn}(x)\Big(\log\big(-\lambda - |x| + (1+\lambda)\sqrt{1+x^2}\big)\mathbf{1}\{|x| < a_\lambda\} + \log(\sqrt{\lambda^2+2\lambda} - \lambda)\mathbf{1}\{|x| \geq a_\lambda\}\Big), \quad (2.9)$$

where $a_\lambda = (\lambda^2 + 2\lambda)^{-1/2}$, and $0 < b_\lambda < a_\lambda$ is such that

$$\log\big(-\lambda + b_\lambda + (1+\lambda)\sqrt{1+b_\lambda^2}\big) = -\log\big(\sqrt{\lambda^2+2\lambda} - \lambda\big).$$

We consider these comparisons because of the following.

Lemma 2. *For any $1 < q \leq 2$, $\lambda > 0$ and for all $x \in \mathbb{R}$, we have for $j = 1, 2$,*

$$-\log(1 - x + (1+\lambda)c_q|x|^q) \leq \varphi_{j,\lambda}(x) \leq \log(1 + x + (1+\lambda)c_q|x|^q).$$

Assuming the settings in Theorem 3, but with $1+\lambda$ replacing $5/4$ in α', $\beta_{n,q,u}$ and the inequality for $\log(1/\delta)/n$ there. Then the result in Theorem 3 holds for $\widehat{\mu}_{\alpha'}$ satisfying $\sum_{i=1}^n \varphi_{j,\lambda}(\alpha'(x_i - \widehat{\mu}_{\alpha'})) = 0$.

Proof of the lemma will be given in Section 6. See Figure 2 for an illustration of the inequalities in the lemma when $\lambda = 0.25$. This lemma suggests that, in light of Theorem 2, $\varphi_{i,0.25}(x)$ should be comparable to $\psi_2(x)$ in performance. And changing the value of λ can have potential improvements. Note that when $\lambda = 0.1$, $\varphi_{2,0.1}(x)$ is very close to the Huber loss's influence function. Hence including $\lambda = 0.1$ in our comparisons essentially includes the Huber loss as well.

For practical implementations, for $\varphi_{j,\lambda}(x)$, we use an α similar to α' in Theorem 3 when solving

$\sum_{i=1}^{n} \varphi_{j,\lambda}(\alpha(x_i - \theta)) = 0$, with 5/4 replaced by $1+\lambda$ and q set to 2:

$$\alpha = \frac{1}{v}\left(\frac{2\log(1/\delta)}{(1+\lambda)n}\right)^{1/2}.$$

The value of v is as in (2.7).

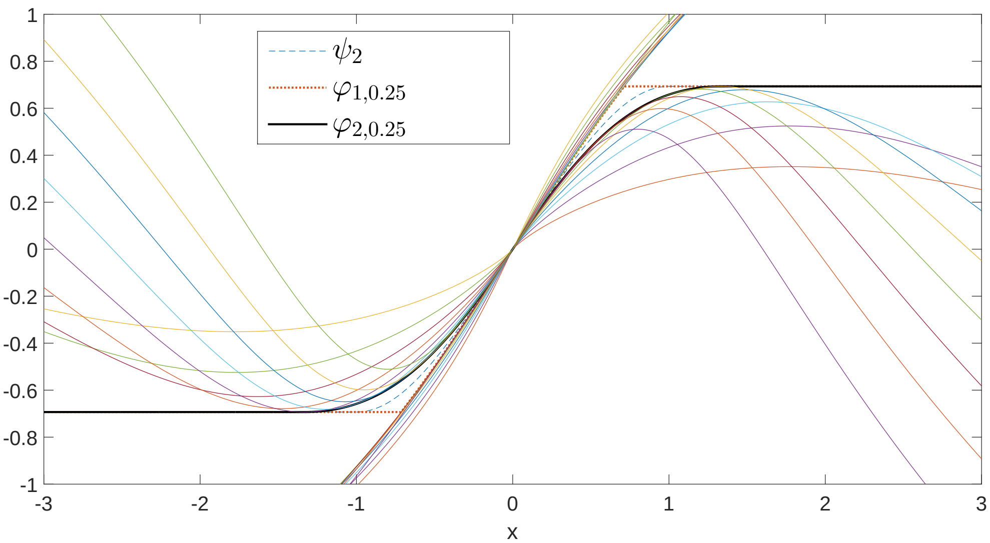

Figure 2: Plot of ψ_2, $\varphi_{1,0.25}$ and $\varphi_{2,0.25}$. The other 18 curves are $\log(1 + x + 1.25c_q|x|^q)$ and the anti-symmetric $-\log(1 - x + 1.25c_q|x|^q)$ (9 of each) for $q = 1.2, 1.3, \ldots, 2$.

3 Shrinkage Covariance Matrix Estimation

It is well-documented that when the dimension p is large compared to the sample size n, the sample covariance matrix performs poorly due to severely biased largest and smallest eigenvalues relative to the true covariance matrix. Random matrix theory has provided theoretical insights into this phenomenon (See Bai and Silverstein (2010) for example). The methods of linear shrinkage (Ledoit and Wolf, 2004) and nonlinear shrinkage (Lam, 2016, Ledoit and Wolf, 2012) are regularization techniques to "correct" these biased eigenvalues in a certain optimal sense, when we do not know any underlying structures of the true covariance matrix $\mathbf{\Sigma}_p$ (see Lam (2020) for a brief overview). However, the aforementioned papers all require the existence of at least 12th order moments for the data. This is a tall order for a lot of applications including finance and econometrics, where for instance financial returns are believed to have not much more than the fourth order moments.

Over the past two decades, a lot of efforts are put into robust estimations, which include robust mean and covariance estimations in particular. These methods require the existence of only second and fourth order moments for mean and covariance estimations respectively. Minsker and Wei (2017) tackle the estimation of high dimensional covariance matrix with heavy-tailed data by using an influence function to limit the contribution of data points that have "large" deviations from the mean, with the same idea modified when using U-statistics in Minsker and Wei (2018). Wegkamp and Zhao (2016) propose using the Kendall's tau for correlation matrix estimation for data following a semiparametric copula model.

One common theme of these papers is that there are no explicit structural assumptions on $\mathbf{\Sigma}_p$ (apart from bounds related to factor models). Unlike shrinkage methods however, they provide no further regularization for tackling the issue of biased eigenvalues when p is large relative to n.

In this section, we bridge the gap between robust covariance estimation and shrinkage methods for estimating a high dimensional covariance matrix, by introducing a shrinkage covariance matrix estimator that is robust to heavy-tailedness. The shrinkage idea is the same as that in Lam (2016), but data allowed can be heavy-tailed, with existence of moments of order only up to $2 + \zeta$ for some $\zeta > 0$. In Section 5, we compare our results to large covariance matrix estimators that are also robust to heavy-tailedness without the need for 4th order moments to exist. For instance, the estimator proposed in Avella-Medina et al. (2018), which has theories developed for sparse $\mathbf{\Sigma}_p$.

3.1 Methodology

Let $\{\mathbf{y}_i\}, i = 1, \ldots, n$ be a random sample of p-dimensional observed vectors. In this paper, we consider $p = p_n$, which can be such that $p/n \to c \in (0, \infty]$. Note that unlike Lam (2016), we include infinity for the ratio p/n, hence we are allowing p to grow even faster than n potentially.

Assume that $E(\mathbf{y}_i) = \boldsymbol{\mu}$ and $\text{var}(\mathbf{y}_i) = \mathbf{\Sigma}_p$, which is our main object of interest in this Section. We allow $\boldsymbol{\mu} \neq 0$, which is a high dimensional nuisance parameter here, and is a significant relaxation to Lam (2016) which assumes $\boldsymbol{\mu} = \mathbf{0}$.

Following Lam (2016), suppose we split the data $\mathbf{Y} = (\mathbf{y}_1, \ldots, \mathbf{y}_n)$ into two independent parts, say $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$, where $\mathbf{Y}_j$ has size $p \times n_j$, $j = 1, 2$, with $n = n_1 + n_2$. Denote for $j = 1, 2$,

$$\widetilde{\mathbf{\Sigma}}_j := \frac{1}{n_j} \sum_{i=1}^{n_j} (\mathbf{y}_{ji} - \bar{\mathbf{y}}_j)(\mathbf{y}_{ji} - \bar{\mathbf{y}}_j)^{\mathrm{T}} = n_j^{-1} \mathbf{Y}_j (\mathbf{I}_p - n_j^{-1} \mathbf{1}_p \mathbf{1}_p^{\mathrm{T}}) \mathbf{Y}_j^{\mathrm{T}},$$

where $\mathbf{I}_p$ is the identity matrix of size p and $\mathbf{1}_p$ is a p-dimensional vector of all ones. We use the notation $\mathbf{Y}_j = (\mathbf{y}_{j1}, \ldots, \mathbf{y}_{jn_j})$, $j = 1, 2$. Consider the nonparametric eigenvalues regularized covariance matrix estimator (NERCOME) in Lam (2016), where

$$\check{\mathbf{\Sigma}}_{n_1} = \mathbf{P}_1 \mathrm{diag}(\mathbf{P}_1 \widetilde{\mathbf{\Sigma}}_2 \mathbf{P}_1) \mathbf{P}_1^{\mathrm{T}}, \tag{3.1}$$

with $\mathbf{P}_1$ being the eigenmatrix of $\widetilde{\mathbf{\Sigma}}_1$, so that $\widetilde{\mathbf{\Sigma}}_1 = \mathbf{P}_1 \mathbf{D}_1 \mathbf{P}_1^{\mathrm{T}}$ with $\mathbf{D}_1$ a diagonal matrix.

Write $\mathbf{P}_1 = (\mathbf{p}_{11}, \ldots, \mathbf{p}_{1p})$. Observe that for $j = 1, \ldots, p$,

$$\mathbf{p}_{1j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1j} = \frac{1}{n_2} \sum_{i=1}^{n_2} (\mathbf{p}_{1j}^{\mathrm{T}} \mathbf{y}_{2i})^2 - (\mathbf{p}_{1j}^{\mathrm{T}} \bar{\mathbf{y}}_2)^2 = \frac{1}{n_2} \sum_{i=1}^{n_2} \left(\mathbf{p}_{1j}^{\mathrm{T}} \mathbf{y}_{2i} - \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbf{p}_{1j}^{\mathrm{T}} \mathbf{y}_{2i} \right)^2.$$

Hence a robust version of $\mathbf{p}_{1j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1j}$ can be

$$\begin{aligned} \widehat{\mathbf{p}_{1j}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1j}} &= \widehat{\theta}_j, \text{ where} \\ \widehat{\theta}_j &= \text{Robust mean of } \{(\mathbf{p}_{1j}^{\mathrm{T}} \mathbf{y}_{2i} - \widehat{\phi}_j)^2\}_{i=1,\ldots,n_2}, \\ \widehat{\phi}_j &= \text{Robust mean of } \{\mathbf{p}_{1j}^{\mathrm{T}} \mathbf{y}_{2i}\}_{i=1,\ldots,n_2}. \end{aligned}$$

Hence a robust version of NERCOME in (3.1) is then

$$\widehat{\mathbf{\Sigma}}_{n_1} = \mathbf{P}_1 \mathrm{diag}(\widehat{\theta}_1, \ldots, \widehat{\theta}_p) \mathbf{P}_1^{\mathrm{T}}. \tag{3.2}$$

The above estimator contains exactly $2p$ robust mean calculations. It avoids the $O(p^2)$ robust mean calculations when we consider robust evaluation of each element in the sample covariance matrix of $\mathbf{Y}$. Also, compared to a plethora of papers using the Kendall rank correlation matrix (a.k.a. Kendall's tau) to recover the Pearson's correlation matrix (see Fan et al. (2019), Xue and Zou (2014), or Wegkamp and Zhao (2016), to name but a few), our estimator has the advantage that the distribution of the data is not limited to the elliptical family.

3.2 Improvement with averaging

Similar to Lam (2016), we can permute the data and construct another estimator using the same splitting scheme for the permuted data. Suppose $\mathbf{Y}^{(j)} = (\mathbf{Y}_1^{(j)}, \mathbf{Y}_2^{(j)})$ is the jth permutation of the data, where $j = 1, \ldots, M$, and M is assumed finite. Then similar to (3.2), we can construct

$$\widehat{\mathbf{\Sigma}}_{n_1}^{(j)} = \mathbf{P}_{1j} \mathrm{diag}(\widehat{\theta}_1^{(j)}, \ldots, \widehat{\theta}_p^{(j)}) \mathbf{P}_{1j}^{\mathrm{T}}, \tag{3.3}$$

where $\mathbf{P}_{1j} = (\mathbf{p}_{11}^{(j)}, \ldots, \mathbf{p}_{1p}^{(j)})$ is the eigenmatrix for the sample covariance matrix

$$\widehat{\mathbf{\Sigma}}_1^{(j)} := n_1^{-1}\mathbf{Y}_1^{(j)}(\mathbf{I}_p - n_1^{-1}\mathbf{1}_p\mathbf{1}_p^{\mathrm{T}})\mathbf{Y}_1^{(j)\mathrm{T}},$$

and for $r = 1, \ldots, p$, the robust mean estimators are defined by

$$\begin{aligned}\widehat{\theta}_r^{(j)} &= \text{Robust mean of } \{(\mathbf{p}_{1r}^{(j)\mathrm{T}}\mathbf{y}_{2i}^{(j)} - \widehat{\phi}_r^{(j)})^2\}_{i=1,\ldots,n_2},\\ \widehat{\phi}_r^{(j)} &= \text{Robust mean of } \{\mathbf{p}_{1r}^{(j)\mathrm{T}}\mathbf{y}_{2i}^{(j)}\}_{i=1,\ldots,n_2}.\end{aligned}$$

The averaged estimator is then defined by

$$\widehat{\mathbf{\Sigma}}_{n_1,M} = \frac{1}{M}\sum_{j=1}^{M}\widehat{\mathbf{\Sigma}}_{n_1}^{(j)}. \tag{3.4}$$

This estimator is more stable than $\widehat{\mathbf{\Sigma}}_{n_1}$ in (3.2), and our empirical experiments are performed using the averaging estimator.

4 Relative Loss to NERCOME

We compare our estimator directly to $\check{\mathbf{\Sigma}}_{n_1}$ (NERCOME) in (3.1), since when the data have enough order of moments exist, $\check{\mathbf{\Sigma}}_{n_1}$ enjoys asymptotic efficiency relative to an ideal estimator (Lam, 2016). For our purpose, we consider the relative loss

$$RL(\widehat{\mathbf{\Sigma}}_{n_1}, \check{\mathbf{\Sigma}}_{n_1}) := \frac{L(\widehat{\mathbf{\Sigma}}_{n_1}, \mathbf{\Sigma}_p)}{L(\check{\mathbf{\Sigma}}_{n_1}, \mathbf{\Sigma}_p)}, \tag{4.1}$$

where we consider the Frobenius loss function

$$L(\widehat{\mathbf{\Sigma}}, \mathbf{\Sigma}_p) := \left\|\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}_p\right\|_F^2, \tag{4.2}$$

or the inverse Stein's loss

$$L(\widehat{\mathbf{\Sigma}}, \mathbf{\Sigma}_p) := \mathrm{tr}(\mathbf{\Sigma}_p\widehat{\mathbf{\Sigma}}^{-1}) - \log\det(\mathbf{\Sigma}_p\widehat{\mathbf{\Sigma}}^{-1}) - p. \tag{4.3}$$

Note that both $\widehat{\mathbf{\Sigma}}_{n_1}$ and $\check{\mathbf{\Sigma}}_{n_1}$ are using the same split location, and the same $\mathbf{Y}_1$ and $\mathbf{Y}_2$ for their constructions. When $RL(\widehat{\mathbf{\Sigma}}_{n_1}, \check{\mathbf{\Sigma}}_{n_1}) \leq 1$, it means that $\widehat{\mathbf{\Sigma}}_{n_1}$ is doing at least as good as $\check{\mathbf{\Sigma}}_{n_1}$, and vice versa.

Before we present our main theorem for this section, we preset more assumptions as follows:

(A1) For each $i = 1, \ldots, n$, we can write $\mathbf{y}_i = \boldsymbol{\mu} + \boldsymbol{\Sigma}_p^{1/2}\mathbf{z}_i$, where each $\mathbf{z}_i$ is a $p \times 1$ random vector with independent components z_{ij}. Each z_{ij} has mean 0 and unit variance. Assume also the existence of ξth order moments for z_{ij}, $2 < \xi \leq 4$.

(A2) The covariance matrix $\boldsymbol{\Sigma}_p$ has eigenvalues uniformly bounded away from 0, and $\|\boldsymbol{\Sigma}_p\| = O(p^\gamma)$, where $\|\cdot\|$ denotes the spectral norm of a square matrix or a vector.

Assumption (A1) is a significant relaxation to Assumption (A1) of Lam (2016), since we only assume the existence of moments up to the ξth order, which can be just larger than 2, instead of requiring 12th order moments to exist. At the same time, (A2) allows $\boldsymbol{\mu}$ to be non-zero and can even have arbitrarily large entries. Moreover, no distributional assumptions like symmetry is imposed, and hence asymmetric distributions are allowed, like mixture normals for example. The range of values of γ allowed is explicitly stated in the following theorem about the relative loss for $\widehat{\boldsymbol{\Sigma}}_{n_1}$.

Theorem 4. *Let condition (A1) and (A2) hold. Assume also that we use the influence functions $\psi_2(x)$, $\varphi_{1,\lambda}(x)$ or $\varphi_{2,\lambda}(x)$ for the robust estimators $\widehat{\theta}_j$ and $\widehat{\phi}_j$ for $j = 1, \ldots, p$, while we set $q = 2$ in the corresponding tuning parameter α' as in Theorem 3 or Lemma 2. Assume also that we use split size $n_1 = \lfloor 0.5n \rfloor$.*

Consider the following two scenarios, with $\widehat{v}_j := \mathrm{var}^{1/2}((\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{21} - \widehat{\phi}_j)^2 | \mathbf{Y}_1)$, and $\{b_{nj}\}$ a double array of bounded constants:

i. $2.4 \leq \xi \leq 4$ *in Assumption (A1). Define $\widehat{u}_j := b_{nj}\widehat{v}_j(\log p/n_2)^{-1/3}$ and we use $\widehat{u} = \max_{1\leq j\leq p}\widehat{u}_j$ in the tuning parameter α' for all $j = 1, \ldots, p$. Assume $p^{6\gamma}\log p/n_2 = o(1)$.*

ii. $2 < \xi < 2.4$ *in (A1), and we use the same $\widehat{u}$ in scenario i. Assume $\gamma = 0$ and $\log p/n_2 = o(1)$.*

Under all the scenarios above, we have $RL(\widehat{\boldsymbol{\Sigma}}_{n_1}, \check{\boldsymbol{\Sigma}}_{n_1}) \leq 1$ almost surely with respect to loss functions (4.2) or (4.3), as long as $\boldsymbol{\Sigma}_p \neq \sigma^2\mathbf{I}_p$. We also have $\widehat{\boldsymbol{\Sigma}}_{n_1}$ almost surely positive definite as $n \to \infty$.

In both scenarios in the theorem, we overestimate the spread parameter by an order of $n^{1/3}/(\log p)^{1/3}$ (see (2.7) for robust mean calculations in practice). We need to do this even when we know $\xi = 4$ (so that our specification $q = 2$ is correct since $q = \xi/2$) because of a technical difficulty involved in controlling the probabilistic behaviour of a term. It means that the best rate for $\|\boldsymbol{\Sigma}_p\|$ to grow is $o(p^{1/6})$ (such that we have $p^{6\gamma}\log p/n_2 = o(1)$) under scenario *i*. Scenario *ii* is when we have very fat-tailed distributions for the data.

Unlike Lam (2016) where they choose the split location to be such that $n_2 \asymp n^{1/2}$ for asymptotic efficiency, in our paper we need n_2 to be of the same order of n (see Section 4.1 for more details). This is because theoretical results for the properties of $\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$ are not established like those in Ledoit and Péché (2011), when in our setting at most 4th order moments exist. Using $n_2 \asymp n^{1/2}$ as opposed to $n_2 \asymp n$ does not leave practical or theoretical advantages anymore even for large p and n.

4.1 Practical implementations for robust NERCOME

Unlike Lam (2016) where they proposed the choice of $n_1 = n - an^{1/2}$ and use a criterion to choose the constant a, we fix n_1 and n_2 to be half the size of n. On top of the point argued in the last paragraph, we actually need n_2 to have the same order as n in Theorem 4, which is a matter of controlling the behaviour of the most extreme data points in $\mathbf{Y}_2$. The choice of $n_1 = \lfloor 0.5n \rfloor$ is a convenient choice which has shown to be very versatile towards different degree of heavy-tailedness in the data distributions. Practically speaking, of course one can still let $n_1 = an$ and treat a as a tuning parameter, using criterion such as that in Lam (2016) to choose a, although at a much more expensive computational cost.

5 Empirical Studies

We start this section with a comparison of mean estimation performance between two M-estimators, followed by a simulation study on robust kurtosis estimation. Finally, we compare our robust NERCOME to other state-of-the-art estimators under different settings, accompanied by an application in ionosphere classification.

5.1 Empirical study of robust mean estimators

We compare our robust mean estimator using the influence function $\varphi_{1,\lambda}$ with $\lambda = 0.4$ defined in (2.8) to the one using the Huber loss in Avella-Medina et al. (2018) (Avella's). Both estimators need an input for q (ϵ in Avella-Medina et al. (2018)), and we assume such knowledge in one setting, but we use $q = 2$ and estimate v using the scheme proposed in (2.7) in another. In both settings, we set $a = 0.1$, $b = 1$ for using the influence function $\varphi_{1,0.4}$ and the same a but $b = 0.5$ for using Huber loss.

We consider $\delta = 2.5\%$ or 5%, sample size $n = 100, 500, 1000$ or 5000, and repeat the experiments 10000 times for each (δ, n) combination. We consider either the $t_{1.51} + 2$ distribution, or the Log-logistic distribution with scale $= 1$ and shape $= 1.51$. The latter posts a challenge to the estimators since the data distribution is skewed. Figure 3 shows the results. First let's assume the knowledge of q_0 (here we know that $q_0 < 1.51$ for both distributions, and we set $q = 1.5$) and v_0, i.e., we compare the left hand side in each of the group boxplots (blue median line). Using $\varphi_{1,\lambda}$ with α' leads to smaller variance for

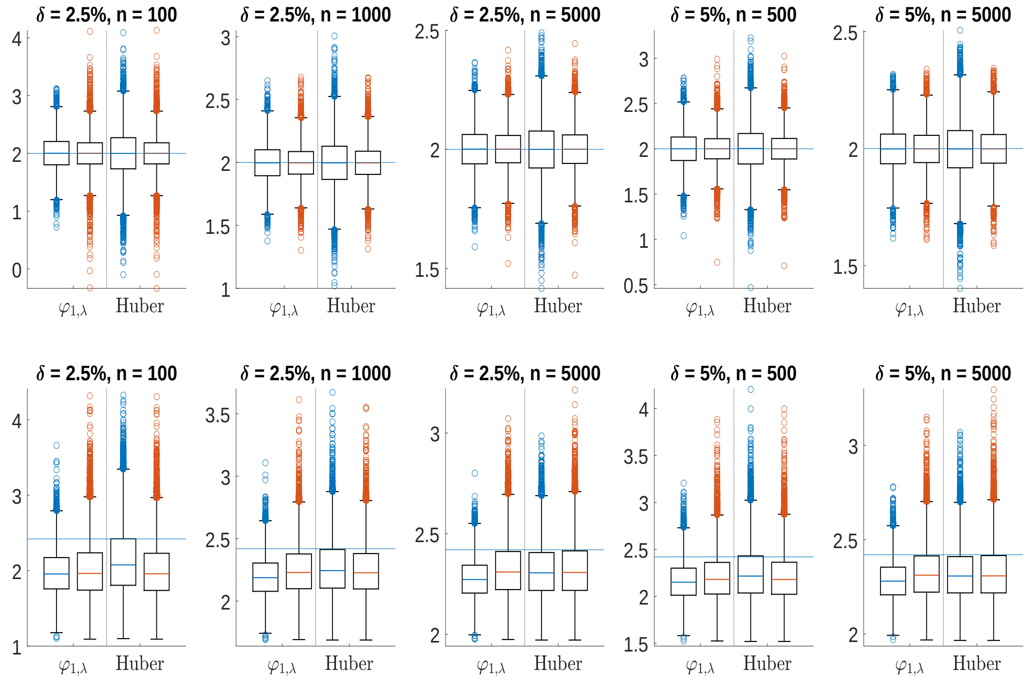

Figure 3: Boxplots of estimates for different methods ($\varphi_{1,\lambda}$: $\lambda = 0.4$, $\widehat{\mu}_{\alpha'}$ using the influence function $\varphi_{1,\lambda}$, α' as in Theorem 3. Huber: use Huber loss, with tuning parameter as suggested in Proposition 6 of Avella-Medina et al. (2018)). *First row*: Data follows $t_{1.51} + 2$ distribution. *Second row*: Data follows Log-logistic(1,1.51). In each of the 10 plots, the blue horizontal line represents the true mean for the data. The left group of two boxplots corresponds to estimates resulted from using $\varphi_{1,\lambda}$, and the right group corresponds to using the Huber loss. Within each group of two boxplots, the left (blue median line) corresponds to using $q = 1.5$ and $v = E^{1/1.5}|x_i - 2|^{1.5}$, while the right (red median line) corresponds to using $q = 2$, and v is estimated using the scheme in (2.7) (For Huber loss, the multiplying factor in v changes from $(0.1n^{1/3} + 1)$ to $(0.1n^{1/3} + 0.5)$). Sample means are calculated but performance is much worse than the two methods and hence is not shown.

the estimator than using Huber loss with H suggested in Avella-Medina et al. (2018), noting that both α' and H are growing at $n^{2/3}$. For symmetric distribution ($t_{1.51} + 2$ in our simulations), there is no bias for both methods. However for skewed distribution (Log-logistic, positively skewed), both methods suffered a downward bias. While the bias is smaller for using Avella's estimator, our estimator has smaller variance.

The picture changes when we use $q = 2$ and estimate v using the scheme in (2.7), i.e., we are comparing the right hand side in each of the group boxplots (red median line). Both estimators are virtually having the same performances, with our estimator having slightly smaller variance when n is large.

For $t_{1.51} + 2$, using $q = 2$ and our scheme in overestimating v by a factor of $n^{1/3}$ in fact leads to even smaller inter quartile range for both methods than when using $q = 1.5$ and v_0. For Avella's, this is because H is larger although the rate of n is still the same. For our method, the rate of $1/\alpha'$ has changed from $n^{2/3}$ to $n^{5/6}$. Both changes imply that more data are allowed into calculating the robust mean without their effects curbed (i.e., within the non-flat region of the influence function) for both methods. For symmetric distribution, it means that inter quartile range can be smaller since not all simulated data sets have a lot of “outliers”.

For Log-logistic distribution however, because it is very positively skewed, allowing more data into the calculations will likely produce more extreme estimates, inflating the variance of the estimator. This is what we see for our method, and when n is large for Avella’s.

We have also carried out simulations under normal distribution for the data. Both methods and the sample mean perform exactly the same, and hence their plots are not shown.

5.2 Kurtosis estimation

The estimation of kurtosis is more sensitive to the presence of extreme outliers than for mean and variance since kurtosis involves higher order moments. Our aim here is to examine the differences between the classical method of using sample kurtosis and an estimator using our robust mean estimation procedure.

We consider data coming from a mixture normal with density $w\phi(-2,4)+(1-w)\phi(0,1)$, where $\phi(\mu,\sigma^2)$ is the density of a Gaussian random variable with mean μ and variance σ^2. We consider 3 different profiles, where $w = 0.01$, 0.05, and 0.1, with the corresponding theoretical kurtosis being 3.97, 5.73, and 6.00. We also consider sample size $n = 50$, 100, 500, 1000, and 5000 to show the finite and large sample behaviour of the estimators, while repeating each setting for 1000 times.

In Figure 4, a prominent feature is the downward bias for both estimators, which goes to 0 as sample size grows. The two estimators have approximately the same median estimates in all cases. However, the inter-quartile ranges of the robust estimates are typically narrower for smaller sample sizes. The exact outlier markers are hidden to preserve an appropriate scale, and the count of outliers (in red) is reported instead beyond the whisker bounds. The sample kurtosis displays an increasing number of outliers when the sample size decreases. On the other hand, robust estimator maintains the number of outliers at a similar level to the case when n=5000. Overall, we can see that our proposed robust estimator can add value in a small sample study on kurtosis.

5.3 Large covariance matrix estimation - comparisons

In this section, we compare NERCOME with robust NERCOME proposed in (3.4) as well as several other alternatives. Specifically, we consider:

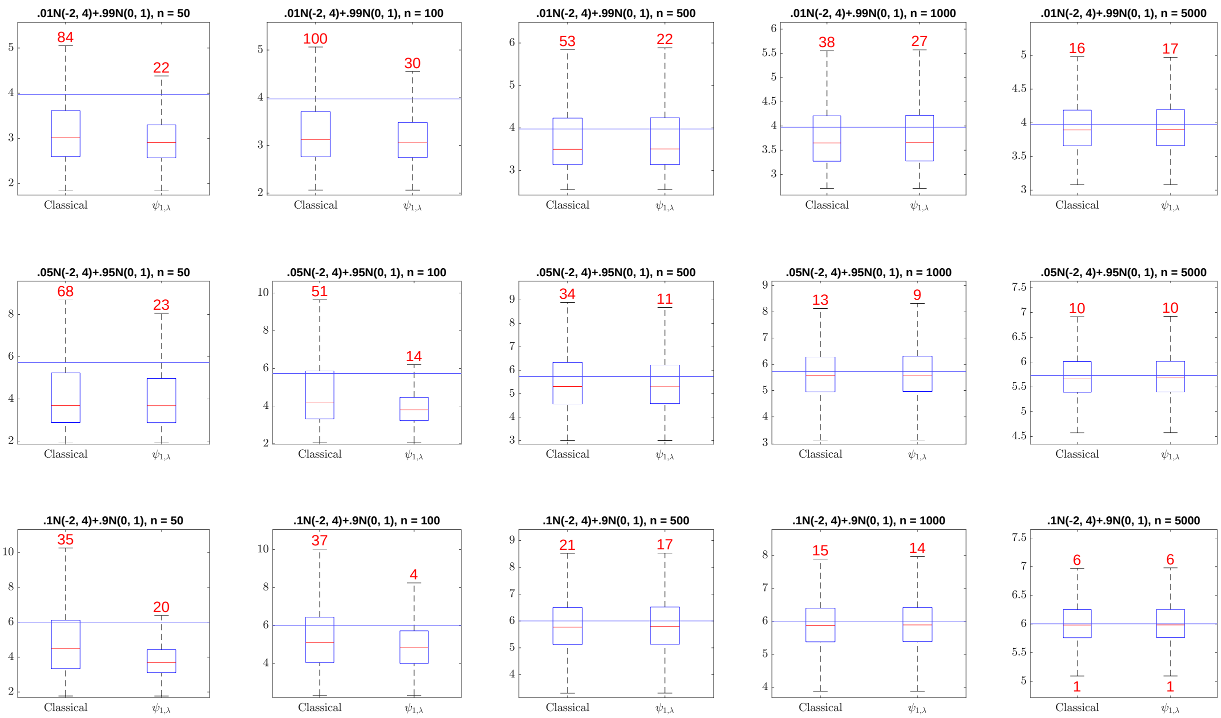

Figure 4: Boxplots of kurtosis estimates using the classical estimator (left in each subplot, sample kurtosis) and using the $\psi_{1,0.25}$ influence function (right in each subplot). We use $\psi_{1,0.25}$ as the influence function to calculate robust mean, variance, and fourth moment for the right-hand plot, with $q = 2$, multiplying factor in v as $2(n/2)^{1/3} - 3$, and 10%-trimmed means as initial estimates. Each kurtosis κ calculated is then corrected for bias using $\frac{n-1}{(n-2)(n-3)}((n+1)\kappa - 3(n-1)) + 3$. Top row: $w = 1\%$. Middle row: $w = 5\%$. Bottom row: $w = 10\%$. Columns from left to right correspond to n=50, 100, 500, 1000, and 5000 respectively. Outlier markers are omitted due to their large scales. We instead provide the counts of outliers next to the bounds of the whiskers. The horizontal blue line in each plot denotes the theoretical kurtosis values.

1) **N**: NERCOME as in Lam (2016), except that we fix the split location at $0.5n$ without searching for optimal split. We produce 30 permutations each time.

2) **RN**: Robust NERCOME as proposed in this paper. We use the influence function of $\psi_{1,0.02}$, $q = 2$, multiplying factor in v as $2(n/2)^{1/3} - 3$, and 10%-trimmed means as initial estimates. Split and permutation setup is the same as in NERCOME. We tried using sample variances for initial estimates of v following e.g., Fan et al. (2017). However, this occasionally produces disastrous outcomes.

3) **HKT**: Robust variances using Huber loss, coupled with Kendall's τ for correlation matrix estimation for reconstruction of the robust covariance matrix (as in Fan et al. (2018)), but with 10%-trimmed estimates in place of sample variances for initial estimate of v_j. Then we replace v_j by $(1 + |\kappa_j|)n^{1/3} \max_j(v_j)$, where κ_j is the trimmed skewness estimate for dimension j, as defined in (2.7). The setup of α follows

Avella-Medina et al. (2018) with $\epsilon = 1$, which is equivalent to setting $q = 2$ for robust NERCOME. No permutation or data splitting is needed.

4) **HKTN**: Same as NERCOME, but $\widetilde{\mathbf{\Sigma}}_2$ is estimated as in HKT using half of the permuted data. The value of v_j is set as $6(1 + |\kappa_j|)n^{1/3} \max_j(v_j)$. Splitting scheme and permutation setup is the same as that in NERCOME.

5) **HT**: Each element of the covariance matrix is estimated using the Huber loss, with 10%-trimmed variances for initial estimates of v_{ij}, which is then replaced by $(1 + |\kappa_{ij}|)n^{1/3} \max_{i,j}(v_{ij})$, where κ_{ij} is the trimmed skewness estimate for element $Y_i \times Y_j$ (Y_i is demeaned). No permutation or split is needed.

The method HKT is just the same as the covariance matrix estimator proposed in Fan et al. (2018), but with v_j set differently to improve its performance. This is also the same in principle to the rank-based estimator described in Avella-Medina et al. (2018) using the Kendall's tau trick. Since we do not assume sparseness or specific structures in $\mathbf{\Sigma}_p$, for fairer comparisons to NERCOME or robust NERCOME, we introduce HKTN - a NERCOME-like estimator which regularize its eigenvalues, but the sample covariance matrix inside (see (3.1)) is now replaced by a robust estimator using the method HKT. The method HT is to do robust estimation on each element of the covariance matrix using the Huber loss function, which is the adaptive Huber estimator described in Avella-Medina et al. (2018). Extra steps are done to project the estimators HKT, HKTN and HT to a positive semi-definite cone using the method proposed in Zhao et al. (2014) to ensure their positive semi-definiteness. NERCOME is proved in Lam (2016) to be positive semi-definite almost surely under light tail assumption, while our proposed robust NERCOME is positive semi-definite by construction. Both do not need the projection treatment.

We demonstrate the performance of these estimators under five profiles for $\mathbf{\Sigma}_p$:

(I) $\mathbf{\Sigma}_p = \mathbf{Q}\mathbf{D}\mathbf{Q}^T$, where $\mathbf{Q}$ is orthogonal and is randomly generated, and $\mathbf{D}$ is diagonal with half of the values being 3 and the other half being 10.

(II) Identity matrix, where $\mathbf{\Sigma}_p = \mathbf{I}_p$.

(III) The data is generated with a factor model structure $\mathbf{y}_t = \mathbf{A}\mathbf{x}_t + \boldsymbol{\epsilon}_t$, where $\mathbf{A}$ has size $p \times 3$ with elements independently generated from $N(0, 2^2)$, $\mathbf{x}_t$'s are independently generated from $N(\mathbf{0}, 2\mathbf{I}_3)$, and the elements in the $\boldsymbol{\epsilon}_t$'s are drawn independently from various (standardised) distributions as specified in (i) - (iii) below.

(IV) Same structure as in (I), however elements in $\mathbf{D}$ is randomly drawn from $U(0, 10p^{1/2})$ instead.

(V) Sparse $\mathbf{\Sigma}_p$ with 20% of its elements being non-zero and a condition number of 10.

Except for factor model, our simulated data set follows $\mathbf{y}_t = \mathbf{\Sigma}_p^{1/2}\mathbf{x}_t + \boldsymbol{\mu}_p$, where $\mathbf{x}_t \in \mathbb{R}^{p \times n}$ has elements independently generated from a chosen standardized marginal distribution from (i) - (iii) below. The mean vector $\boldsymbol{\mu}_p$ has elements drawn from t_2. In the factor model in (III), it is the elements in $\boldsymbol{\epsilon}_t$ that are drawn from different standardized marginal distributions independently.

We consider 6 marginal distributions in 3 categories, including:

(i) symmetric distributions: $N(0,1)$ and $t_{2.2}$;

(ii) asymmetric distributions: Weibull (1, 0.25) and Lognormal$(1, 1.2^2)$;

(iii) mixed distributions: a mixture of normal distributions with $N(-5, 4^2)$ and $N(1,1)$ equally weighted, and a mixture of Student-t distributions of $t_{2.2}$ with mean -5 and t_5 with mean 1, also equally weighted.

For each combination of marginal distribution and $\boldsymbol{\Sigma}_p$ profile, we consider (p,n) = (200, 100), (200, 500), (400, 100), and (400, 500), and run 300 simulations for each of the these (p,n) combinations. We gauge the performance using Frobenius loss (FL) defined in (4.2), and the inverse Stein's loss (SL) defined in (4.3). When both $p = 200$ and $p = 400$ are involved for comparisons, the losses are divided by p^2 to make results comparable.

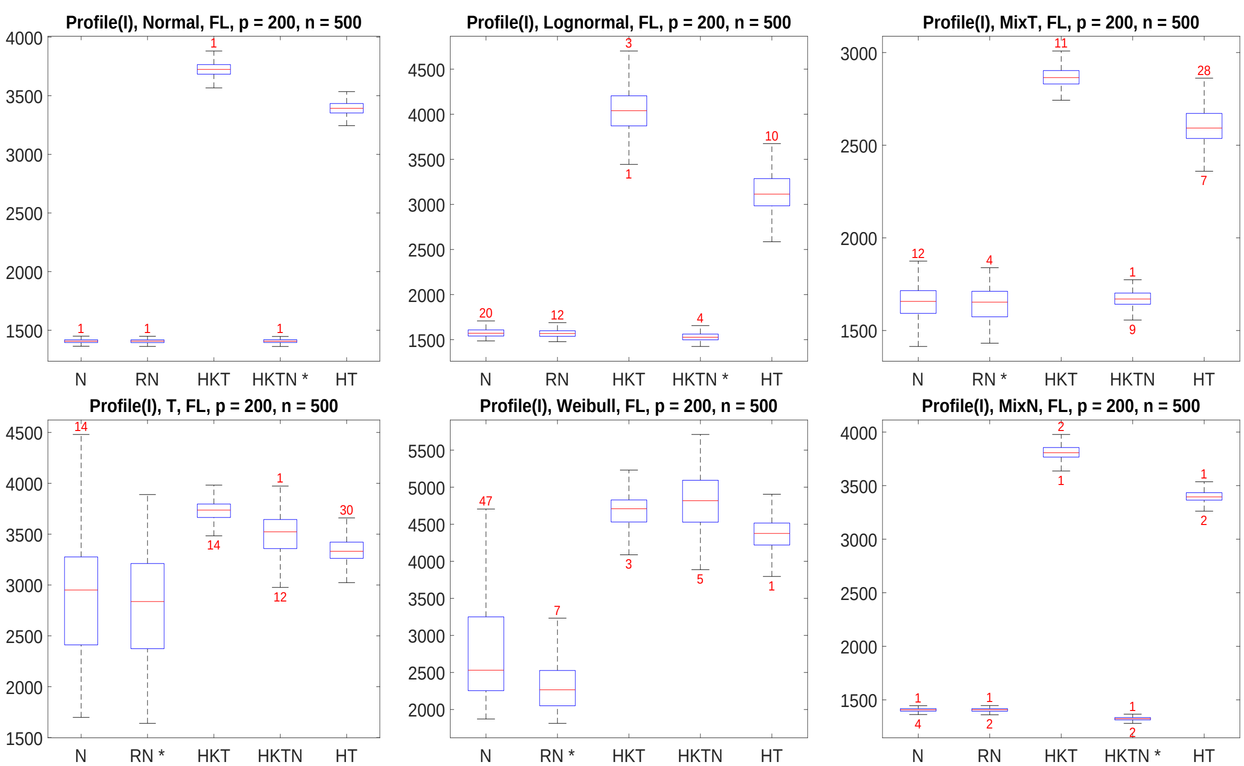

Figure 5: Frobenius loss for profile(I) with $p = 200, n = 500$. *Left column*: symmetric distributions under (i). *Middle Column*: asymmetric distributions under (ii). *Right column*: mixed distributions under (iii). The numbers in red are the number of outliers beyond the upper or lower whiskers. A * means the method has the smallest median Frobenius error.

Figure 5 presents an overall view of all the five estimators studied in terms of the Frobenius loss under profile (I). Here we illustrate with $p = 200$ and $n = 500$ only, because in other cases and particularly when $p > n$, the performance gap among the methods are even bigger, and are all in favour of N, RN, and HKTN. We observe similar performance gap in favour of N, RN and HKTN for other profiles of $\mathbf{\Sigma}_p$ under different (p, n) combinations. Hence eigenvalue shrinkage does add value towards Frobenius loss minimization, and in subsequent analysis we do not include HKT and HT to simplify comparisons.

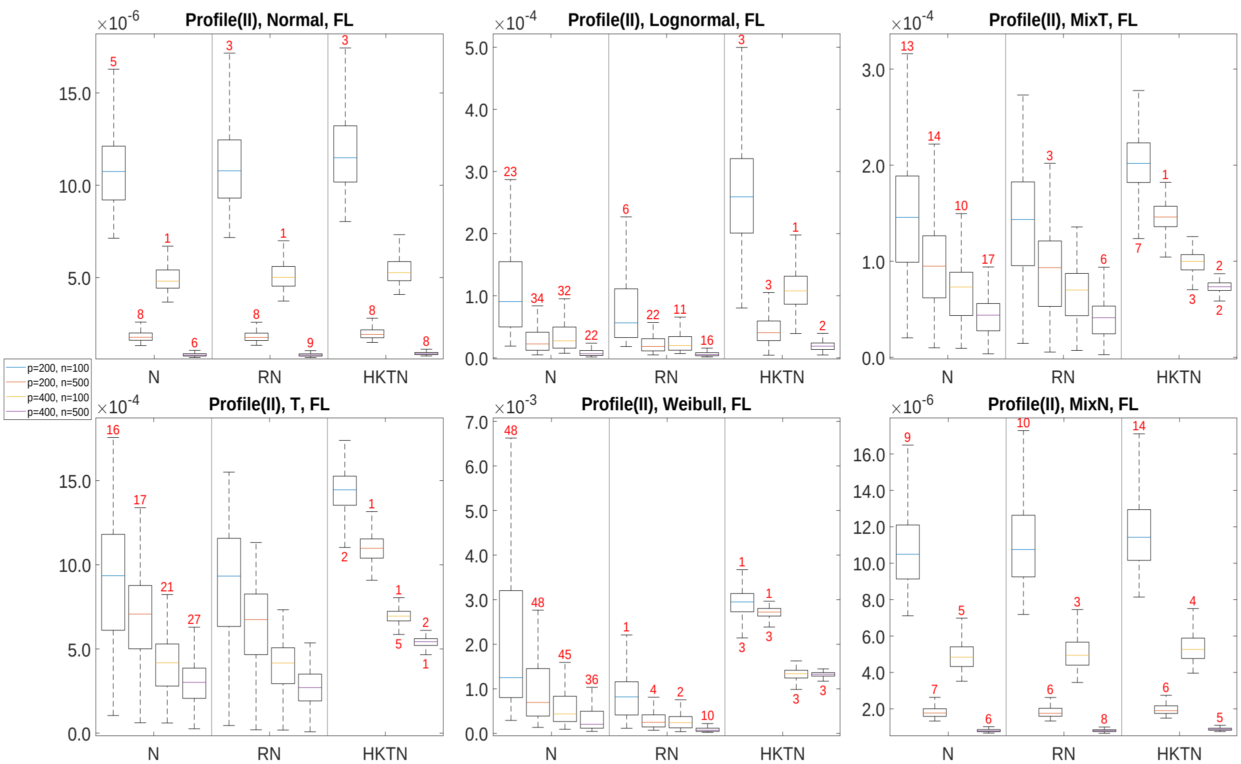

Figure 6: Frobenius loss divided by p^2 for profile(II). *Left column*: symmetric distributions under (i). *Middle Column*: asymmetric distributions under (ii). *Right column*: mixed distributions under (iii). Each subplot presents a set of four boxplots (corresponding to four combinations of p and n) for each of the three estimators (N, RN, and HKTN) from left to right. The numbers in red are the number of outliers beyond the upper or lower whiskers.

Figure 6 shows the Frobenius loss divided by p^2 (i.e., the average elementwise sum of squares loss) for profile (II) where $\mathbf{\Sigma}_p = \mathbf{I}_p$. Under normal distribution, all three methods perform similarly with the robust versions slightly worse than NERCOME. For $t_{2.2}$, with rather limited existence of moments, both RN and HKTN have effectively limited the number of outliers to a few, while NERCOME produces around 20 outliers out of 300 simulations. However, RN maintains similar levels of median losses as in NERCOME, while HKTN's are in general much higher. Under asymmetric distributional settings (middle column), NERCOME produces even more outliers than in $t_{2.2}$. Meanwhile, RN not only reduces the number of outliers but also decreases the median and variance of losses. Despite the fact that Kendall's τ in theory does not work with asymmetric distributions, HKTN produces the smallest number of outliers among the three, although its overall performance is inferior. For mixed normal distribution (right column, bottom subplot), the picture looks very similar to that under normal distribution. For mixed t distribution (right column, top subplot), the performance of all methods are better than when under $t_{2.2}$ despite the similarity of the plots. Finally, all the distributional settings see consistent decrease in the average sum of squares loss when either n or p increases.

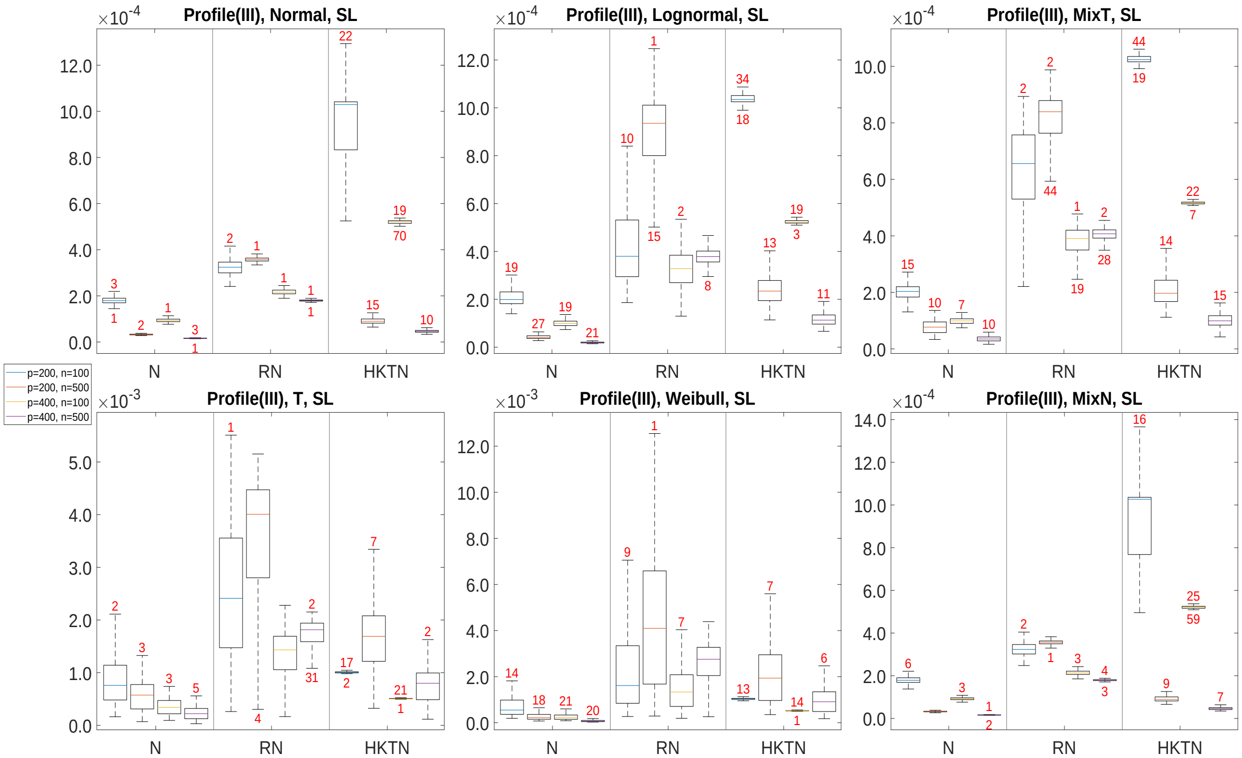

Figure 7: Inverse Stein's loss divided by p^2 for profile(III). Layouts are the same as in Figure 6.

Next we look into the results on data generated from a three-factor model in Figure 7. We do this comparisons to explore mainly how robust NERCOME compares to NERCOME although we do not have the corresponding theory on a factor model for robust NERCOME. Here we use the inverse Stein's loss divided by p^2, i.e., the average elementwise Stein's loss, since it is equally bad for the three methods using Frobenius loss. Proven to be optimal with factor models in Lam (2016) under light-tailed factors and errors only, NERCOME still stands out under all current distributional settings. Between RN and HKTN, the latter seem to outperform and particularly so when $n > p$.

For profile (IV) and (V), we use the relative loss defined in (4.1) and gauge the performance of robust NERCOME and HKTN relative to NERCOME. A ratio lower than 1 (referenced by the blue horizontal line) indicates the corresponding robust estimator has a better performance than NERCOME, and vice versa.

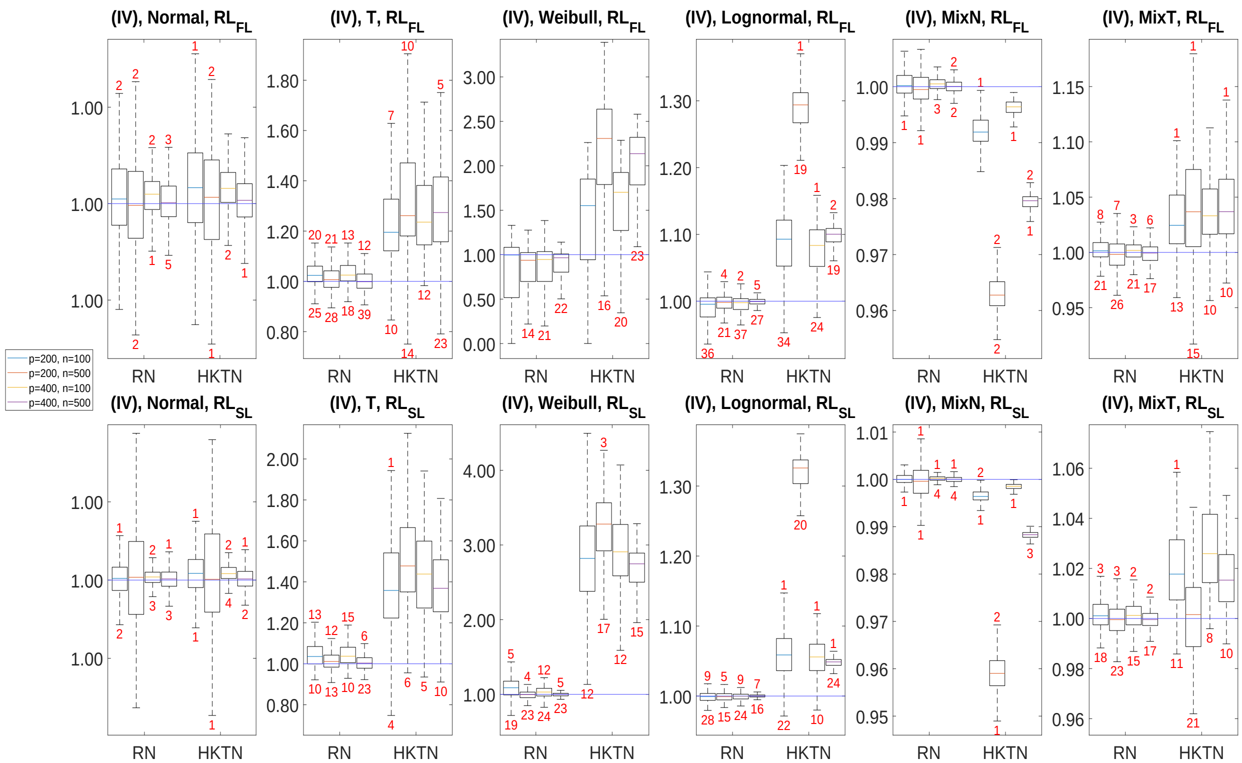

Figure 8: Relative loss defined in (4.1) for RN and HKTN relative to NERCOME with respect to Frobenius loss (upper row) and Stein's loss (lower row) for profile(IV). Each column corresponds to a marginal distribution. Each subplot presents a set of four boxplots (corresponding to four combinations of p and n) for RN (left) and HKTN (right). The numbers in red are the number of outliers beyond the upper or lower whiskers.

Figure 8 shows the results under profile (IV). It is clear that robust NERCOME outperforms HKTN by quite some margins, particularly for heavy tailed $t_{2.2}$ and highly skewed Weibull distributions. For mixed normal distribution, HKTN performs slightly better, but percentage-wise the difference between robust NERCOME and HKTN is not huge at all. The median for robust NERCOME is typically not too far from 1 and it seems to balance efficiency and robustness well, evidenced by a relatively small number of outliers above 1 as compared to the number of outliers below 1.

Figure 9 shows profile (V) with sparse $\mathbf{\Sigma}_p$. For normal distribution, HKTN in general gives wider inter-quartile ranges than that for robust NERCOME. Across different distributions, HKTN has mixed performance. It has again performed a bit better under mixed normal distribution, given all of its boxes are below one, although not too far from it. However it worked out poorly under $t_{2.2}$ and Weibull, with all of its boxes quite far above from 1. For lognormal, the boxes of HKTN are below 1 only when $p < n$. Robust NERCOME performs similarly as in profile(IV) with its median around 1 and number of outliers on the upside mostly smaller than on the downside. Additionally, the range of relative loss often shrinks with increasing p and n, for both loss functions.

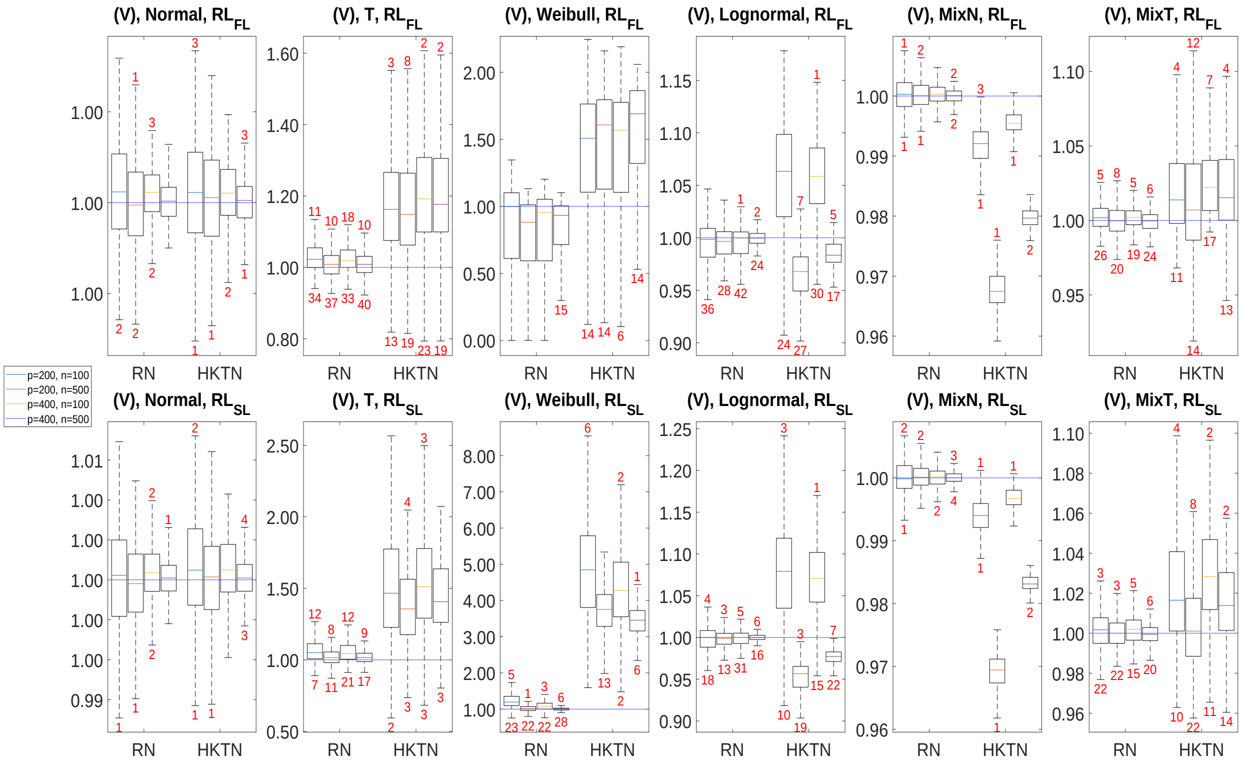

Figure 9: Relative loss with respect to Frobenius loss (upper row) and Stein's loss (lower row) for profile(V). Layout is the same as in figure 8.

5.4 Ionosphere classification study

The Goose Bay radar system employed a phased array of 16 high-frequency antennas to scan the ionosphere for structures. The dataset consists of 351 received signals, of which 225 "good" ones display evidence of some type of structure and 126 "bad" ones do not. Each signal contains 32 non-zero attributes after being processed with autocorrelation functions on 17 pulse numbers recorded. The dataset can be downloaded from UCI machine learning repository (https://archive.ics.uci.edu/ml/datasets/ionosphere). The task here is to tell whether a signal is "good" or "bad" and it has been studied by Ghosh et al. (2020) under the framework of generalised quadratic discriminant analysis (GQDA), where efforts were made to robustify GQDA with sample mean and sample covariance estimators replaced by robust estimators in the literature. Likewise, we focus on the out-of-sample misclassification error rates among the sample estimator, NERCOME and the robust NERCOME. As in Ghosh et al. (2020), we pose a slightly more challenging setup with 70% of the data in the training set and contaminate 20% of it by flipping their original classes. We repeat the experiment for 1000 times with a random partition each time. The results are summarised in Figure 10. The results show overall lower median misclassification error rates for NERCOME (8.57%) and robust NERCOME (7.62%) compared to using sample estimators (12.38%). Between NERCOME and robust NERCOME, the latter has slightly lower median and 75-th percentile values than the former. Though not immediately clear from the boxplot, it is worth pointing out that among the 46 outliers on the top end in R-NERCOME, only 4 are above the upper adjacent values of NERCOME.

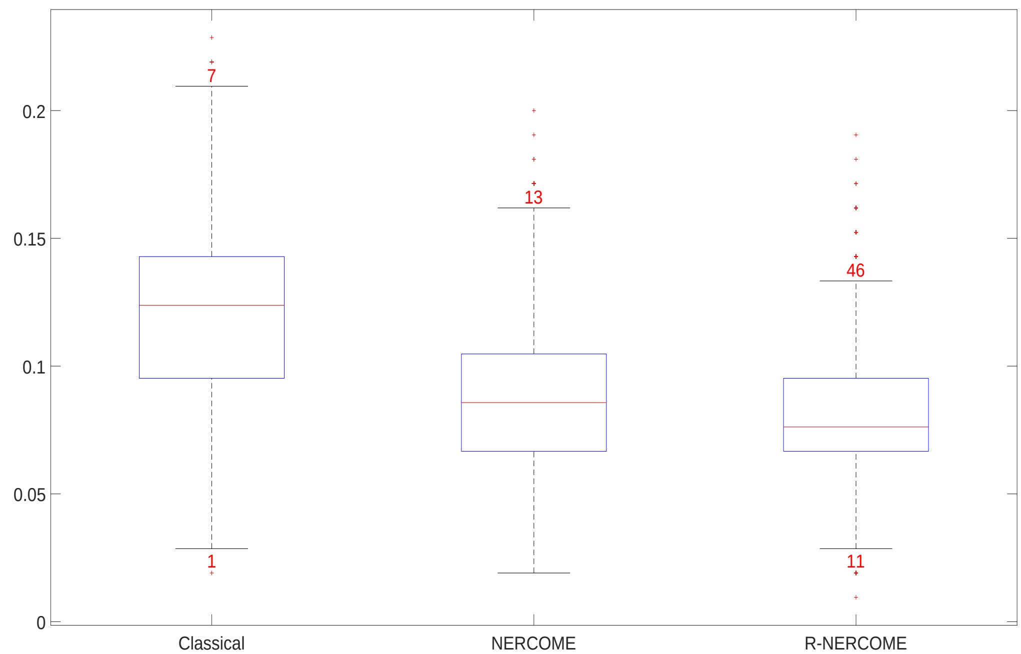

Figure 10: Boxplots of misclassification error rates from GQDA with inputs generated by three methods. Classical: use sample mean and sample covariance. NERCOME: use sample mean and NERCOME for covariance. R-NERCOME: use robust mean and robust NERCOME for covariance. For R-NERCOME, the same configuration applies to both mean and covariance estimation, where the influence function $\psi_{2,0.4}$ is used with $q = 2$ and v is set to have multiplying factors $0.5n^{1/3}$ and $0.1n^{1/3}$ for mean and covariance estimation respectively. The numbers in red are the number of outliers beyond the upper or lower whiskers. Some red '+' markers have multiple outliers behind.

6 Proofs

In this section, we present Lemma 3 and its proofs. We also provide proofs of Theorem 1, Lemma 1, Theorem 2, Theorem 3, Lemma 2 and Theorem 4.

Lemma 3. *One of the narrowest possible non-decreasing curve ψ satisfying (2.2) for a fixed $1 < q \leq 2$ is given by ψ_q in (2.3).*

Proof of Lemma 3. First of all, note that the curves

$$g_+ := \log(1 + x + c|x|^q), \quad g_- := -\log(1 - x + c|x|^q)$$

are anti-symmetric around 0 (i.e., $g_+(-x) = -g_-(x)$ for each x). In order that $g_+(x) \geq g_-(x)$ for all x, we solve $\log(1 + x + c|x|^q) \geq -\log(1 - x + c|x|^q)$, which can be simplified to

$$c \geq \frac{(1 + x^2)^{1/2} - 1}{|x|^q} =: k(x), \quad x \in \mathbb{R}.$$

Observe that $k(x)$ is continuous and differentiable on $\mathbb{R}/\{0\}$, and is symmetric around 0. Also,

$$\lim_{x \to \pm\infty} k(x) = 0, \quad \lim_{x \to 0} k(x) = \begin{cases} 0, & 1 < q < 2; \\ 1/2, & q = 2. \end{cases}$$

Hence the only stationary point of $k(x)$ for $x \geq 0$ at $x_q = (2/q - 1)^{1/2}/(1 - 1/q)$ represents a maximum, and hence the minimum of c allowed is

$$c = k(x_q) =: c_q = q^{-q/2}(2 - q)^{1-q/2}(q - 1)^{q-1}.$$

With $c = c_q$, the only maximum point of g_- is at $b_q = (qc_q)^{1/(1-q)}$ (and hence the only minimum of g_+ is at $-b_q$). For one of the narrowest possible non-decreasing ψ satisfying (2.2), we must have that $\psi(x)$ is a constant at $g_+(-b_q) =: -a_q$ for $x \leq -b_q$, and at $g_-(b_q) = a_q$ for $x \geq b_q$, and that it is increasing and satisfies (2.2) for $|x| < b_q$. It is easy to verify then ψ_q in (2.3) satisfies all of these, since it follows exactly g_+ for $-b_q \leq x \leq 0$ and g_- for $0 \leq x \leq b_q$. □

Proof of Theorem 1. The proof follows the same spirit as the proof of Theorem 5 in Fan et al. (2017) and Proposition 2.4 in Catoni (2012). Define $g(\theta) := \frac{1}{n\alpha}\sum_{i=1}^n \psi_q(\alpha(x_i - \theta))$. We have $g(\widehat{\mu}_\alpha) = 0$ by definition.

Since by construction ψ_q satisfies (2.2), by independence of the x_i's, we have

$$\begin{aligned} E\big(\exp(n\alpha g(\theta))\big) &= E^n\big(\exp(\psi_q(\alpha(x_1 - \theta)))\big) \leq \big(1 + \alpha(\mu - \theta) + c_q\alpha^q E|x_1 - \theta|^q\big)^n \\ &\leq \exp\big\{n\alpha(\mu - \theta) + nc_q\alpha^q d_q\big(v^q + |\mu - \theta|^q\big)\big\}, \end{aligned} \tag{6.1}$$

where we used the C_r-inequality to have $E|y_1 - \theta|^q \leq 2^{q-1}(E|y_1 - \mu|^q + |\mu - \theta|^q) \leq 2^{q-1}(v^q + |\mu - \theta|^q)$ for

$1 < q \leq 2$, and the fact that $v \geq E^{1/q}|x_1 - \mu|^q$. Similarly, we have

$$E\big(\exp(-n\alpha g(\theta))\big) \leq \exp\big\{-n\alpha(\mu-\theta) + nc_q\alpha^q d_q(v^q + |\mu-\theta|^q)\big\}. \tag{6.2}$$

Define

$$\begin{aligned} B_+(\theta) &:= (\mu-\theta) + c_q d_q \alpha^{q-1}(v^q + |\mu-\theta|^q) + \frac{\log(1/\delta)}{n\alpha}, \\ B_-(\theta) &:= (\mu-\theta) - c_q d_q \alpha^{q-1}(v^q + |\mu-\theta|^q) - \frac{\log(1/\delta)}{n\alpha}. \end{aligned}$$

Hence using Markov's inequality,

$$P(g(\theta) > B_+(\theta)) \leq \frac{\exp\big\{n\alpha(\mu-\theta) + nc_q\alpha^q d_q\big(v^q + |\mu-\theta|^q\big)\big\}}{\exp\big\{n\alpha(\mu-\theta) + nc_q\alpha^q d_q\big(v^q + |\mu-\theta|^q\big)\big\}\exp(\log(1/\delta))} = \delta,$$

and similarly, $P(g(\theta) < B_-(\theta)) \leq \delta$. Hence

$$P(B_-(\theta) \leq g(\theta) \leq B_+(\theta)) \geq 1 - 2\delta. \tag{6.3}$$

Define $\beta_n := d_q c_q \alpha^{q-1} v^q + \frac{\log(1/\delta)}{n\alpha}$. Then for $1 < k \leq 2$, we have

$$\begin{aligned} B_+(\mu+\beta_n) &= d_q c_q \alpha^{q-1}\beta_n^q > 0, \\ B_+(\mu+k\beta_n) &= k^q d_q c_q \alpha^{q-1}\beta_n^q - (k-1)\beta_n = \beta_n(k^q d_q c_q \alpha^{q-1}\beta_n^{q-1} - (k-1)). \end{aligned} \tag{6.4}$$

Similarly, we have

$$\begin{aligned} B_-(\mu-\beta_n) &= -d_q c_q \alpha^{q-1}\beta_n^q < 0, \\ B_-(\mu-k\beta_n) &= -\beta_n(k^q d_q c_q \alpha^{q-1}\beta_n^{q-1} - (k-1)) = -B_+(\mu+k\beta_n). \end{aligned} \tag{6.5}$$

Let $\theta+$ be the root of $B_+(\theta) = 0$ and θ_- be the root of $B_-(\theta) = 0$. If we can assume that $B_+(\mu+k\beta_n) < 0$ (so that $B_-(\mu-k\beta_n) > 0$), then we have

$$\mu + \beta_n \leq \theta_+ \leq \mu + k\beta_n, \quad \mu - k\beta_n \leq \theta_- \leq \mu - \beta_n.$$

With the above, since $g(\theta)$ is non-increasing in θ, under the event $\{B_-(\theta) \leq g(\theta) \leq B_+(\theta)\}$, we have

$$\mu - k\beta_n \leq \theta_- \leq \widehat{\mu}_\alpha \leq \theta_+ \leq \mu + k\beta_n,$$

which implies that $|\widehat{\mu}_\alpha - \mu| \le k\beta_n$. Hence

$$1 - 2\delta \le P(B_-(\theta) \le g(\theta) \le B_+(\theta)) \le P(|\widehat{\mu}_\alpha - \mu| \le k\beta_n), \tag{6.6}$$

which is equivalent to the result of the theorem if we can show $B_+(\mu + k\beta_n) < 0$, $B_-(\mu - k\beta_n) > 0$, and that

$$\beta_n = \frac{q d_q^{1/q} c_q^{1/q} v}{(q-1)^{1-1/q}} \left(\frac{\log(1/\delta)}{n}\right)^{1-1/q}. \tag{6.7}$$

To this end, note that β_n is itself a function of α. From (6.6), we see that $2k\beta_n$ is the width of the confidence interval (with confidence level at least $1 - 2\delta$) for μ. Hence we want to choose an α to minimize β_n.

Simple calculus shows that at

$$\alpha = \left(\frac{1}{d_q c_q (q-1)}\right)^{1/q} \frac{1}{v} \left(\frac{\log(1/\delta)}{n}\right)^{1/q}, \tag{6.8}$$

β_n is minimized at (6.7). It remains to show $B_+(\mu + k\beta_n) < 0$ (which implies $B_-(\mu - k\beta_n) > 0$). From (6.4), $B_+(\mu + k\beta_n) < 0$ is equivalent to

$$(\alpha\beta_n)^{q-1} < \frac{k-1}{k^q d_q c_q}.$$

But from the value of α in (6.8) and the corresponding minimized value of β_n in (6.7), we have that $\alpha\beta_n = q\log(1/\delta)/((q-1)n)$. Hence the above implies that

$$\frac{\log(1/\delta)}{n} < \frac{q-1}{q} \left(\frac{k-1}{k^q d_q c_q}\right)^{1/(q-1)},$$

which is our requirement stated in the theorem. This completes the proof of the theorem. □

Proof of Lemma 1. We first assume that $\psi(x) \le g_{+,h}(x) := \log(1 + x + c_{q,h}|x|^q)$ for all x. Then we also have $\psi(-x) \le g_{+,h}(-x)$, which implies

$$\psi(x) = -\psi(-x) \ge -g_{+,h}(-x) = -\log(1 - x + c_{q,h}|x|^q) =: g_{-,h}(x).$$

And with $g_{-,h}(x) \le \psi(x) \le g_{+,h}(x)$ for all x, if $\widehat{\mu}_{\alpha,h}$ is the solution to $\sum_{i=1}^n \psi(\alpha(x_i - \theta)) = 0$, then we can follow exactly the steps in the proof of Theorem 1 to arrive at the result and requirement in Theorem 1 where $\widehat{\mu}_\alpha$ and c_q there are replaced by $\widehat{\mu}_{\alpha,h}$ and $c_{q,h}$ respectively. Hence it remains to show $\psi(x) \le g_{+,h}(x)$ for all x.

To this end, note that $g_{+,h}$ is continuous and is differentiable for $x \in \mathbb{R}$. By simple calculus, $g_{+,h}$ has a global minimum at $x_0 = -(qc_{q,h})^{1/(1-q)}$. Since $g_{+,h}$ is increasing for $x \geq 1$, we have for $x \geq 1$,

$$\psi(x) = 1 = \log(1 + 1 + (e-2)) \leq g_{+,h}(1) \leq g_{+,h}(x).$$

For $x \leq -1$, it is sufficient to prove that $g_{+,h}(x_0) \geq -1 = \psi(x)$, since $g_{+,h}$ has a global minimum at x_0. By the definition of $c_{q,h}$, for $1 < q \leq 2$,

$$\begin{aligned} c_{q,h} &\geq q^{-q}(q-1)^{q-1}(1-e^{-1})^{1-q}, \text{ which is equivalent to} \\ c_{q,h}^{-1/(q-1)}&[q^{-1/(q-1)}(q^{-1}-1)] \geq e^{-1} - 1. \end{aligned}$$

Rearranging, we have

$$1 - (qc_{q,h})^{1/(1-q)} + c_{q,h}(qc_{q,h})^{q/(1-q)} \geq e^{-1},$$

which is equivalent to $g_{+,h}(x_0) \geq -1$.

Finally, for $|x| \leq 1$, consider $k(x) := (e^x - 1 - x)|x|^{-q}$, which has

$$\lim_{x \to 0} k(x) = \begin{cases} 0, & 1 < q < 2; \\ 1/2, & q = 2. \end{cases}$$

For $x > 0$, we have

$$k(x) - k(-x) = |x|^{-q}(e^x - e^{-x} - 2x) = 2|x|^{-q} \sum_{s \geq 1} x^{2s+1}/(2s+1)! > 0,$$

and hence we only need to consider $k(x)$ for $x > 0$ when considering the maximum of $k(x)$ in $[-1, 1]$. Since for $x > 0$ and $1 < q \leq 2$, we have

$$k'(x) = \sum_{s \geq 2} \frac{s-q}{s!} x^{s-1-q} > 0,$$

the maximum of $k(x)$ is either $k(0)$ or $k(1)$ on $[-1, 1]$. But $k(1) = e - 2 \leq c_{q,h}$, and hence we have for $-1 \leq x \leq 1$,

$$1 + x + c_{q,h}|x|^q \geq e^x,$$

which is true for $x = 0$ as well, and is equivalent to $g_{+,h}(x) \geq \psi(x)$. This completes the proof of the lemma.
□

Proof of Theorem 2. With the inequalities in the statement of the theorem, we can follow exactly the same steps as in the proof of Theorem 1 to arrive at the same result and requirement as in Theorem 1, with q and c_q there (both in the statements in the theorem and its proof) replaced by q_0 and $5/4c_{q_0}$ respectively. Hence it remains to show the inequalities.

It is sufficient to prove the right hand side inequality, since with this, and using that ψ is anti-symmetric around 0, we have

$$\psi(x) = -\psi(-x) \geq -\log\biggl(1 - x + \frac{5}{4}c_q|x|^q\biggr).$$

Define $g_q(x) := \log\bigl(1 + x + \frac{5}{4}c_q|x|^q\bigr)$.

Case I: $x \geq 1$. We have $\psi_2(x) = \log(2)$. It is obvious that g_q is increasing over $x > 1$, and hence

$$\psi_2(x) = \log(2) \leq \log(2 + 5c_q/4) = g_q(1) \leq g_q(x).$$

Case II: $0 \leq x < 1$. We have $\psi_2(x) = -\log(1 - x + x^2/2)$. Consider

$$k(x) := \biggl(1 + x + \frac{5}{4}c_q x^q\biggr)(1 - x + x^2/2).$$

We need to show $k(x) \geq 1$, which is equivalent to $\psi_2(x) \leq g_q(x)$.

First expand $k(x)$, then rearrange terms, and multiply both sides by 2 to get that $k(x) \geq 1$ is equivalent to $\frac{5}{4}c_q x^q(x^2 - 2x + 2) - x^2 + x^3 \geq 0$. Using the fact that $0 \leq x < 1$ and $c_q \geq \frac{2}{5}$ (see Case III below), we have the left hand side $\geq \frac{1}{2}x^2(x^2 - 2x + 2) - x^2 + x^3 = x^4/2 \geq 0$.

Case III: $-1 \leq x < 0$. We have $\psi_2(x) = \log(1 + x + x^2/2)$. If we can show that

$$\frac{x^2}{2} \leq \frac{5}{4}c_q|x|^q,$$

then the proof of this part would be completed. To this end, it would be sufficient to show that $c_q \geq 0.4$ since $-1 \leq x < 0$ and $1 < q \leq 2$. Consider the derivative of c_q with respect to q,

$$(c_q)' = c_q\log\biggl(\frac{q-1}{\sqrt{2q - q^2}}\biggr),$$

which is strictly increasing and maps bijectively from $(1, 2]$ to the whole real line. The root of $(c_q)' = 0$ is at $q = 1 + 1/\sqrt{2}$ in $(1, 2]$, and the minimum value of c_q is at $\sqrt{2} - 1 > 0.4$.

Case IV: $x < -1$. We have $\psi_2(x) = -\log(2)$. Hence we need to show

$$\frac{1}{2} \leq 1 + x + \frac{5}{4}c_q(-x)^q =: h_q(x).$$

Consider

$$h'_q(x) = 1 - \frac{5}{4} q c_q (-x)^{q-1} = 0$$

at $x_0 = -(4/(5qc_q))^{1/(q-1)}$. If $x_0 > -1$, i.e., $qc_q > 0.8$, then $h_q(x)$ has a minimum at $x = -1$ on $(-\infty, -1]$, and $h_q(-1) = 5c_q/4 \geq 5(\sqrt{2} - 1)/4 > 1/2$.

Now consider $qc_q \leq 0.8$, so that $x_0 \leq -1$. Then

$$h_q(x_0) = 1 - |x_0| + \frac{5}{4} c_q |x_0|^q = 1 - |x_0| + \frac{1}{q}|x_0| \geq \frac{1}{2},$$

if we can show that $|x_0| \leq q/(2(q-1))$, which is equivalent to

$$\frac{4}{5} \leq qc_q \cdot \frac{q^{q-1}}{2^{q-1}(q-1)^{q-1}} = \frac{q^{q/2}(2-q)^{1-q/2}}{2^{q-1}} =: g(q).$$

Now

$$g'(q) = g(q) \log\left(\frac{1}{2}\sqrt{\frac{q}{2-q}}\right) =: g(q) f(q),$$

with $f(q)$ strictly increasing and maps from $(1, 2]$ bijectively to $(-\log(2), \infty]$. Also, $f(q) = 0$ when $q = 1.6$, with the minimum value of $g(q)$ at $g(1.6) = 0.8$. Finally, we indeed have $1.6c_{1.6} = 0.6732 < 0.8$. This completes the proof of the Theorem. $\square$

Proof of Theorem 3. From Theorem 1 and Theorem 2, we know that the ideal tuning parameter to use is (with q replaced by q_0 and c_q replaced by $5c_{q_0}/4$)

$$\alpha_0 = \left(\frac{4}{5c_{q_0} d_{q_0}(q_0 - 1)}\right)^{1/q_0} \frac{1}{v_0} \left(\frac{\log(1/\delta)}{n}\right)^{1/q_0},$$

where $v_0 = E^{1/q_0}|x_1 - \mu|^{q_0}$.

Following the very same steps as in the proof of Theorem 1, we can define

$$\begin{aligned}
\beta_n(\alpha) &:= \frac{5}{4} c_{q_0} \alpha^{q_0 - 1} d_{q_0} v_0^{q_0} + \frac{\log(1/\delta)}{n\alpha}, \\
B_+(\theta) &:= (\mu - \theta) + \frac{5}{4} c_{q_0} \alpha^{q_0 - 1} d_{q_0} (v_0^{q_0} + |\mu - \theta|^{q_0}) + \frac{\log(1/\delta)}{n\alpha}, \\
B_-(\theta) &:= (\mu - \theta) - \frac{5}{4} c_{q_0} \alpha^{q_0 - 1} d_{q_0} (v_0^{q_0} + |\mu - \theta|^{q_0}) - \frac{\log(1/\delta)}{n\alpha},
\end{aligned}$$

so that for $1 < k \leq 2$, if $B_+(\mu + k\beta_n(\alpha)) < 0$, we can have

$$P(|\widehat{\mu}_\alpha - \mu| \geq k\beta_n(\alpha)) \leq 2\delta,$$

where $\widehat{\mu}_\alpha$ is the solution to $\sum_{i=1}^n \psi_2(\alpha(x_i-\theta))=0$. If we replace α by

$$\alpha' = \left(\frac{4}{5c_q d_q(q-1)}\right)^{1/q} \frac{1}{u} \left(\frac{\log(1/\delta)}{n}\right)^{1/q},$$

then noting $1/q = 1/q_0 - \eta$,

$$\begin{aligned}
\beta_n(\alpha') &= \frac{5}{4} c_{q_0} d_{q_0} v_0 \left(\frac{v_0}{u}\right)^{q_0-1} \left(\frac{4}{5d_q c_q(q-1)}\right)^{(q_0-1)/q} \left(\frac{\log(1/\delta)}{n}\right)^{(q_0-1)/q} \\
&\quad + \left(\frac{5d_q c_q(q-1)}{4}\right)^{1/q} u \left(\frac{\log(1/\delta)}{n}\right)^{1-1/q} \\
&= \left(\frac{5(q-1)d_q c_q}{4}\right)^{1/q} \left(\frac{\log(1/\delta)}{n}\right)^{1-1/q_0} \\
&\quad \cdot \left\{ \frac{5}{4} c_{q_0} d_{q_0} v_0 \left(\frac{v_0}{u}\right)^{q_0-1} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \left(\frac{\log(1/\delta)}{n}\right)^{-\eta(q_0-1)} + u \left(\frac{\log(1/\delta)}{n}\right)^{\eta} \right\} = \beta_{n,q,u}.
\end{aligned}$$

Finally, $B_+(\mu + k\beta_{n,q,u}) < 0$ is equivalent to

$$(1-k)\beta_{n,q,u} + \frac{5}{4} c_{q_0} \alpha'^{q_0-1} d_{q_0} k^{q_0} \beta_{n,q,u}^{q_0} = \beta_{n,q,u} \left(\frac{5k^{q_0}}{4} c_{q_0} d_{q_0} (\alpha' \beta_{n,q,u})^{q_0-1} - (k-1) \right) < 0,$$

which is equivalent to

$$\alpha' \beta_{n,q,u} < \left(\frac{4(k-1)}{5k^{q_0} c_{q_0} d_{q_0}}\right)^{1/(q_0-1)}.$$

Since by simple algebra,

$$\alpha' \beta_{n,q,u} = \frac{5}{4} c_{q_0} d_{q_0} \left(\frac{v_0}{u}\right)^{q_0} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \left(\frac{\log(1/\delta)}{n}\right)^{1-\eta q_0} + \frac{\log(1/\delta)}{n},$$

we need

$$\frac{\log(1/\delta)}{n} < \left(\frac{4(k-1)}{5k^{q_0} c_{q_0} d_{q_0}}\right)^{1/(q_0-1)} \left(1 + \frac{5}{4} c_{q_0} d_{q_0} \left(\frac{v_0}{u}\right)^{q_0} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \left(\frac{\log(1/\delta)}{n}\right)^{-\eta q_0} \right)^{-1}.$$

A sufficient condition is then

$$\frac{\log(1/\delta)}{n} < \left(\frac{4(k-1)}{5k^{q_0} c_{q_0} d_{q_0}}\right)^{1/(q_0-1)} \left(\frac{\log(1/\delta)}{n}\right)^{\eta q_0} \left(1 + \frac{5}{4} c_{q_0} d_{q_0} \left(\frac{v_0}{u}\right)^{q_0} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \right)^{-1},$$

which is equivalent to

$$\frac{\log(1/\delta)}{n} < \left(\frac{4(k-1)}{5k^{q_0} c_{q_0} d_{q_0}}\right)^{q/(q_0(q_0-1))} \left(1 + \frac{5}{4} c_{q_0} d_{q_0} \left(\frac{v_0}{u}\right)^{q_0} \left(\frac{4}{5d_q c_q(q-1)}\right)^{q_0/q} \right)^{-q/q_0},$$

since $q_0/q = 1 - \eta q_0$. This completes the proof of the theorem. $\square$

Proof of Lemma 2. With the inequalities in the statement of the theorem, we can follow exactly the same steps as in the proof of Theorem 3 to arrive at the same result and requirement as in Theorem 3, with $5/4$ replaced by $1+\lambda$ in α', $\beta_{n,q,u}$ and the inequality for $\log(1/\delta)/n$. It remains to show the inequalities.

It is sufficient to prove the right hand side inequality only, since then we can use the anti-symmetric property of $\varphi_{i,\lambda}$ around 0 to obtain

$$\varphi_{i,\lambda}(x) = -\varphi_{i,\lambda}(-x) \geq -\log(1 - x + (1+\lambda)c_q|x|^q).$$

From the proof of Lemma 3, we know that $c_q \geq ((1+x^2)^{1/2}-1)/|x|^q$ for all $x \in \mathbb{R}$, $1 < q \leq 2$ and $\lambda > 0$. Hence we must also have

$$(1+\lambda)c_q \geq \frac{(1+x^2)^{1/2}-1}{|x|^q}$$

for all $x \in \mathbb{R}$, $1 < q \leq 2$ and $\lambda > 0$. This translates to

$$-\log(1 - x + (1+\lambda)c_q|x|^q) \leq \log(1 + x + (1+\lambda)c_q|x|^q), \quad x \in \mathbb{R},\ 1 < q \leq 2,\ \lambda > 0.$$

Our next task is to show that for a fix $x \in \mathbb{R}$,

$$\min_{1<q\leq 2}(\log(1 + x + (1+\lambda)c_q|x|^q)) = \log(-\lambda + x + (1+\lambda)\sqrt{1+x^2}), \tag{6.9}$$

$$\max_{1<q\leq 2}(-\log(1 - x + (1+\lambda)c_q|x|^q)) = -\log(-\lambda - x + (1+\lambda)\sqrt{1+x^2}). \tag{6.10}$$

To this end, define $g_{q,\lambda}(x) := \log(1 + x + (1+\lambda)c_q|x|^q)$. Consider

$$\begin{aligned}\frac{dg_{q,\lambda}(x)}{dq} &= \exp\left(-g_{q,\lambda}(x)\right) \cdot \frac{d}{dq}\left((1+\lambda)|x|^q q^{-q/2}(q-1)^{q-1}(2-q)^{1-q/2}\right) \\ &= (1+\lambda)\exp\left(-g_{q,\lambda}(x)\right)|x|^q c_q \log\left(\frac{|x|(q-1)}{\sqrt{q(2-q)}}\right).\end{aligned}$$

Setting the above derivative to 0, we get $q_x := 1 + (1+x^2)^{-1/2}$. This is a minimum since the second derivative is

$$\frac{dg_{q,\lambda}(x)}{dq} = -\left(\frac{dg_{q,\lambda}(x)}{dq}\right)^2 + \frac{dg_{q,\lambda}(x)}{dq}\log\left(\frac{|x|(q-1)}{\sqrt{q(2-q)}}\right) + \frac{(1+\lambda)\exp\left(-g_{q,\lambda}(x)\right)|x|^q c_q}{q(q-1)(2-q)} > 0$$

at q_x. Hence we have

$$\min_{1<q\leq 2} g_{q,\lambda}(x) = g_{q_x,\lambda}(x) = \log(-\lambda + x + (1+\lambda)\sqrt{1+x^2}),$$

which is (6.9). Then

$$\max_{1<q\leq 2}(-g_{q,\lambda}(-x)) = -\min_{1<q\leq 2} g_{q,\lambda}(-x) = -g_{q_{-x},\lambda}(-x) = -\log(-\lambda - x + (1+\lambda)\sqrt{1+x^2}),$$

which is (6.10). Since $q_x = q_{-x}$, we must also have

$$-\log(-\lambda - x + (1+\lambda)\sqrt{1+x^2}) \leq \log(-\lambda + x + (1+\lambda)\sqrt{1+x^2}) =: g_\lambda(x), \quad x \in \mathbb{R}. \tag{6.11}$$

Simple calculus on $g_\lambda(x)$ shows that the global minimum of g_λ is at $x = -(\lambda^2+2\lambda)^{-1/2} = -a_\lambda$, with $g_\lambda(-a_\lambda) = \log(\sqrt{\lambda^2+2\lambda} - \lambda)$. Then for $x \leq -a_\lambda$, we have by construction,

$$\varphi_{2,\lambda}(x) = g_\lambda(-a_\lambda) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

We also have, for $x \leq -b_\lambda$,

$$\varphi_{1,\lambda}(x) = g_\lambda(-a_\lambda) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

Consider $-a_\lambda < x \leq 0$. Then

$$\varphi_{2,\lambda}(x) = g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

For $-b_\lambda < x \leq 0$, by (6.11),

$$\varphi_{1,\lambda}(x) = -g_\lambda(-x) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

Consider $0 < x \leq a_\lambda$, then by (6.11),

$$\varphi_{2,\lambda}(x) = -g_\lambda(-x) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

For $0 < x \leq b_\lambda$,

$$\varphi_{1,\lambda}(x) = g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2,\ \lambda > 0.$$

Finally, consider $x > a_\lambda$. Since $-g_\lambda(-x)$ has a global maximum at a_λ, with $-g(-a_\lambda) = -\log(\sqrt{\lambda^2+2\lambda} - \lambda)$, we have by (6.11),

$$\varphi_{2,\lambda}(x) = -g_\lambda(-a_\lambda) \leq g_\lambda(a_\lambda) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2, \ \lambda > 0,$$

where the second last inequality used the fact that $g_\lambda(x)$ is increasing for $x > 0$. We also have, for $x > b_\lambda$,

$$\varphi_{1,\lambda}(x) = -g_\lambda(-a_\lambda) = g_\lambda(b_\lambda) \leq g_\lambda(x) \leq g_{q,\lambda}(x), \quad 1 < q \leq 2, \ \lambda > 0,$$

where the second equality is true by the construction of $\varphi_{1,\lambda}(x)$, and the second last inequality used the fact that g_λ is increasing for $x > 0$. This completes the proof of the lemma. □

Proof of Theorem 4. Consider the Frobenius loss in (4.2) first. Define $\widehat{\mathbf{\Theta}} := \mathrm{diag}(\widehat{\theta}_1, \ldots, \widehat{\theta}_p)$. Then

$$\begin{aligned} p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{n_1} - \mathbf{\Sigma}_p\big\|_F^2 &= p^{-1}\big\|\mathbf{P}_1\widehat{\mathbf{\Theta}}\mathbf{P}_1^{\mathrm{T}} - \mathbf{\Sigma}_p\big\|_F^2 = p^{-1}\big\|[\widehat{\mathbf{\Theta}} - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)] + [\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1]\big\|_F^2 \\ &= p^{-1}\big\|\widehat{\mathbf{\Theta}} - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)\big\|_F^2 + p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2. \end{aligned}$$

At the same time, we also have

$$p^{-1}\big\|\check{\mathbf{\Sigma}}_{n_1} - \mathbf{\Sigma}_p\big\|_F^2 = p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1) - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)\big\|_F^2 + p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2.$$

But with $\mathbf{\Sigma}_p \neq \sigma^2\mathbf{I}_p$ and with 12th order moments exist for the data, the proof of Lemma S.4 in the supplement of Lam (2016) shows that

$$p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2 \xrightarrow{a.s.} a_c > 0$$

under $p/n \to c > 0$ (and so $p/n_1 \to 2c$), where a_c is a constant depending on c. Hence in our settings with relaxed moments assumptions, we have $p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2 \neq 0$ almost surely (may actually diverge). With this, we then have

$$\begin{aligned} &RL(\widehat{\mathbf{\Sigma}}_{n_1}, \check{\mathbf{\Sigma}}_{n_1}) = \frac{p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{n_1} - \mathbf{\Sigma}_p\big\|_F^2}{p^{-1}\big\|\check{\mathbf{\Sigma}}_{n_1} - \mathbf{\Sigma}_p\big\|_F^2} \\ &= \left(\frac{p^{-1}\big\|\widehat{\mathbf{\Theta}} - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)\big\|_F^2}{p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2} + 1\right)\left(\frac{p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1) - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)\big\|_F^2}{p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1\big\|_F^2} + 1\right)^{-1}, \end{aligned}$$

which is less than or equal to 1 almost surely if we can prove that

$$p^{-1}\big\|\widehat{\mathbf{\Theta}} - \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_1)\big\|_F^2 \xrightarrow{a.s.} 0. \tag{6.12}$$

To this end, define $\widehat{v}_j := E^{1/q_0}(|(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2 - E((\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2|\mathbf{Y}_1)|^{q_0}|\mathbf{Y}_1)$ where $q_0 = \xi/2$, $\widehat{u}_j := b_{nj}\widehat{v}_j(\log\ p/n_2)^{-1/3}$, $\widehat{u} := \max_{1\leq j\leq p}\widehat{u}_j$. The sequence $\{b_{nj}\}$ is a double array of constants uniformly bounded away from 0 and infinity. Using $q = 2$, define

$$\widehat{\alpha} := \left(\frac{2}{1+\lambda}\right)^{1/2}\frac{1}{\widehat{u}}\left(\frac{3\log p}{n_2}\right)^{1/2}. \tag{6.13}$$

Then $\widehat{\theta}_j$ is the solution to $\widehat{g}_j(\theta) := \frac{1}{n\widehat{\alpha}}\sum_{i=1}^{n_2}\psi(\widehat{\alpha}((\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2 - \theta)) = 0$.

Define $w_{ji} := (\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu}))^2$, and $m_j := E(w_{ji}|\mathbf{Y}_1) = \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j}$, $v_j := E^{1/q_0}(|w_{ji} - m_j|^{q_0}|\mathbf{Y}_1)$. Hereafter we also use $E_{\mathbf{Y}_1}(\cdot)$ and $P_{\mathbf{Y}_1}(\cdot)$ to denote expectation and probability given $\mathbf{Y}_1$.

Since by construction the influence functions we consider are all Lipschitz continuous, we have

$$\begin{aligned}
n_2\widehat{\alpha}\widehat{g}_j(\theta) &= \sum_{i=1}^{n_2}\psi\left\{\widehat{\alpha}(w_{ji} - \theta) + 2\widehat{\alpha}\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})(\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu} - \widehat{\phi}_j) + \widehat{\alpha}(\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu})^2\right\} \\
&\leq \sum_{i=1}^{n_2}\psi(\widehat{\alpha}(w_{ji} - \theta)) + C_\psi\widehat{\alpha}\sum_{i=1}^{n_2}(2|\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| + |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|)|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|,
\end{aligned}$$

where C_ψ is the Lipschitz constant. By a result in Lemma 4, on an almost sure set $\mathcal{M}$, for large enough n (and hence n_2),

$$\exp\left(C_\psi\widehat{\alpha}\sum_{i=1}^{n_2}(2|\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| + |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|)|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|\right) \leq C_p := p^C, \tag{6.14}$$

where $C > 0$ is a constant for each $j = 1, \ldots, p$. Then following the same idea of proof as in Theorem 1, for large enough n (and hence large enough n_2), using the result of Lemma 2 (for $\varphi_{j,\lambda}$) or Theorem 2 (for ψ_2, with $\lambda = 0.25$),

$$\begin{aligned}
E_{\mathbf{Y}_1}(\exp(n_2\widehat{\alpha}\widehat{g}_j(\theta))) &\leq C_pE_{\mathbf{Y}_1}^{n_2}(1 + \widehat{\alpha}(w_{ji} - \theta) + (1+\lambda)c_{q_0}\widehat{\alpha}^{q_0}|w_{ji} - \theta|^{q_0}) \\
&\leq C_p\exp\left\{n_2\widehat{\alpha}(m_j - \theta) + n_2(1+\lambda)c_{q_0}\widehat{\alpha}^{q_0}d_{q_0}(v_j^{q_0} + |m_j - \theta|^{q_0})\right\}.
\end{aligned} \tag{6.15}$$

Similarly, using

$$n_2\widehat{\alpha}\widehat{g}_j(\theta) \geq \sum_{i=1}^{n_2}\psi(\widehat{\alpha}(w_{ji} - \theta)) - C_\psi\widehat{\alpha}\sum_{i=1}^{n_2}(2|\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| + |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|)|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|,$$

when n is large enough, for the exact same C_p,

$$E_{\mathbf{Y}_1}(\exp(-n_2\widehat{\alpha}\widehat{g}_j(\theta))) \leq C_p\exp\left\{-n_2\widehat{\alpha}(m_j - \theta) + n_2(1+\lambda)c_{q_0}\widehat{\alpha}^{q_0}d_{q_0}(v_j^{q_0} + |m_j - \theta|^{q_0})\right\}. \tag{6.16}$$

Define

$$
\begin{aligned}
B_{j+}(\theta) &:= (m_j-\theta)+(1+\lambda)c_{q_0}d_{q_0}\widehat{\alpha}^{q_0-1}(v_j^{q_0}+|m_j-\theta|^{q_0})+\frac{\log(C_p/\delta)}{n_2\widehat{\alpha}},\\
B_{j-}(\theta) &:= (m_j-\theta)-(1+\lambda)c_{q_0}d_{q_0}\widehat{\alpha}^{q_0-1}(v_j^{q_0}+|m_j-\theta|^{q_0})-\frac{\log(C_p/\delta)}{n_2\widehat{\alpha}}.
\end{aligned}
$$

Then using the two inequalities (6.15) and (6.16), we have for large enough n,

$$
P_{\mathbf{Y}_1}(B_{j-}(\theta)\le \widehat{g}_j(\theta)\le B_{j+}(\theta))\ge 1-2\delta.
$$

Now define

$$
\beta_{n_2}^{(j)} := (1+\lambda)d_{q_0}c_{q_0}\widehat{\alpha}^{q_0-1}v_j^{q_0}+\frac{\log(C_p/\delta)}{n_2\widehat{\alpha}}.
$$

Then we can show easily that

$$
\begin{aligned}
&B_{j+}(m_j+\beta_{n_2}^{(j)})>0,\quad B_{j-}(m_j-\beta_{n_2}^{(j)})<0,\\
&B_{j+}(m_j+k\beta_{n_2}^{(j)})=\beta_{n_2}^{(j)}\{(1+\lambda)c_{q_0}d_{q_0}k^{q_0}\widehat{\alpha}^{q_0-1}(\beta_{n_2}^{(j)})^{q_0-1}-(k-1)\}=-B_{j-}(m_j-k\beta_{n_2}^{(j)}).
\end{aligned}
$$

Putting $\delta=p^{-3}$, we can easily show that $B_{j+}(m_j+k\beta_{n_2}^{(j)})<0$ is equivalent to

$$
(1+\lambda)^{1-q_0/2}c_{q_0}d_{q_0}6^{q_0/2}C'\left(\frac{\log p}{n_2}\right)^{5q_0/6}+\frac{C\log p+3\log p}{n_2}<\left(\frac{k-1}{(1+\lambda)c_{q_0}d_{q_0}k^{q_0}}\right)^{1/(q_0-1)}, \tag{6.17}
$$

if we can show that

$$
\frac{\max_j v_j^{q_0}}{\max_j b_{nj}\widehat{v}_j^{q_0}}\le C'
$$

for a constant $C'>0$ for large enough n, which is exactly one of the result of Lemma 4. Hence for large enough n (and so n_2), we do have $B_{j+}(m_j+k\beta_{n_2}^{(j)})<0$ and $B_{j-}(m_j-k\beta_{n_2}^{(j)})>0$ from (6.17) since $p/n_2\to 2c>0$. Hence,

$$
P_{\mathbf{Y}_1}(|\widehat{\theta}_j-m_j|\ge k\beta_{n_2}^{(j)})\le 2p^{-3}. \tag{6.18}
$$

This implies that, for large enough n,

$$
P_{\mathbf{Y}_1}\left(p^{-1}\sum_{j=1}^{p}(\widehat{\theta}_j-m_j)^2\ge p^{-1}k^2\sum_{j=1}^{p}(\beta_{n_2}^{(j)})^2\right)\le 2p^{-2}.
$$

Since $\sum_{n_2\ge 1}2p^{-2}<\infty$ for $p/n\to c>0$ and $n_2=0.5n$, the Borel-Cantelli's lemma implies that

$p^{-1}\sum_{j=1}^{p}(\widehat{\theta}_j - m_j)^2 \overset{a.s.}{\to} 0$, which is (6.12), if we can show that $p^{-1}\sum_{j=1}^{p}(\beta_{n_2}^{(j)})^2 = o(1)$. But

$$
\begin{aligned}
p^{-1}\sum_{j=1}^{p}(\beta_{n_2}^{(j)})^2 &= \max_j(b_{nj}^2\widehat{v}_j^2)\cdot p^{-1}\sum_{j=1}^{p}\left\{(1+\lambda)d_{q_0}c_{q_0}\left(\frac{2}{1+\lambda}\right)^{(q_0-1)/2}\frac{v_j^{q_0}}{\max_j(b_{nj}\widehat{v}_j)^{q_0}}\left(\frac{3\log p}{n_2}\right)^{5(q_0-1)/6}\right.\\
&\qquad\left.+\left(\frac{C\log p+3\log p}{n_2}\right)\left(\frac{1+\lambda}{6}\right)^{1/2}\left(\frac{\log p}{n_2}\right)^{-5/6}\right\}^2\\
&\le \text{Constant}\cdot\max_j(\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j})^2\cdot p^{-1}\sum_{j=1}^{p}\left(\left(\frac{\log p}{n_2}\right)^{5(q_0-1)/6}+\left(\frac{\log p}{n_2}\right)^{1/6}\right)^2\\
&= \begin{cases} O\left(\left\|\boldsymbol{\Sigma}_p\right\|^2\left(\frac{\log p}{n_2}\right)^{1/3}\right), & 1.2 < q_0 \le 2;\\ O\left(\left\|\boldsymbol{\Sigma}_p\right\|^2\left(\frac{\log p}{n_2}\right)^{5(q_0-1)/3}\right), & 1 < q_0 \le 1.2,\end{cases}
\end{aligned}
\tag{6.19}
$$

where the second last inequality used Lemma 4. Since $\xi = 2q_0$, and $\left\|\boldsymbol{\Sigma}_p\right\| = O(p^{\gamma})$, the above 2 cases represent scenarios *i* and *ii* respectively, and all of them are going to 0 by assumptions in the Theorem as $n \to \infty$. This completes the proof of (6.12), and hence the treatment for the Frobenius loss.

For the inverse Stein's loss L, define $f(x) := x - 1 - \log(x)$ which is non-negative for $x > 0$, with a global minimum 0 at $x = 1$. Also, define $\boldsymbol{\Sigma}_{\text{Ideal},n_1} := \mathbf{P}_1\text{diag}(\mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1)\mathbf{P}_1^{\mathrm{T}}$. Note that the inverse Stein's loss is always non-negative for non-negative definite matrices. Hence in particular,

$$
\begin{aligned}
L(\boldsymbol{\Sigma}_{\text{Ideal},n_1},\boldsymbol{\Sigma}_p) &= \text{tr}(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1)\mathbf{P}_1^{\mathrm{T}}) - p - \log\det(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1)\mathbf{P}_1^{\mathrm{T}})\\
&= -\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1}) \ge 0.
\end{aligned}
\tag{6.20}
$$

Then we have

$$
\begin{aligned}
RL(\widehat{\boldsymbol{\Sigma}}_{n_1},\check{\boldsymbol{\Sigma}}_{n_1}) &= \frac{\text{tr}(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}(\widehat{\theta}_1^{-1},\ldots,\widehat{\theta}_p^{-1})\mathbf{P}_1^{\mathrm{T}}) - p - \log\det(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}(\widehat{\theta}_1^{-1},\ldots,\widehat{\theta}_p^{-1})\mathbf{P}_1^{\mathrm{T}})}{\text{tr}(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{P}_1)\mathbf{P}_1^{\mathrm{T}}) - p - \log\det(\boldsymbol{\Sigma}_p\mathbf{P}_1\text{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{P}_1)\mathbf{P}_1^{\mathrm{T}})}\\
&= \frac{p^{-1}\sum_{j=1}^{p}\left(\frac{\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j}}{\widehat{\theta}_j} - 1 - \log\left(\frac{\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j}}{\widehat{\theta}_j}\right)\right) - p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1})}{p^{-1}\sum_{j=1}^{p}\left(\frac{\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j}}{\mathbf{p}_{1j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1j}} - 1 - \log\left(\frac{\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j}}{\mathbf{p}_{1j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1j}}\right)\right) - p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1})}.
\end{aligned}
$$

If $\boldsymbol{\Sigma}_p \ne \sigma^2\mathbf{I}_p$ and 12th order moments exist for the data, then the proof of Lemma S.4 in the supplement of Lam (2016) shows that

$$
-p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1}) \overset{a.s.}{\to} b_c > 0
$$

under $p/n \to c > 0$ (and hence $p/n_1 \to 2c$), where b_c is a constant depending on c. Hence in our setting with relaxed moments assumptions, we have $-p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1}) > 0$ almost surely (can actually diverge). With this, and the fact that $f(x) \geq 0$ for $x > 0$, we then have

$$RL(\widehat{\boldsymbol{\Sigma}}_{n_1}, \check{\boldsymbol{\Sigma}}_{n_1}) = \frac{p^{-1}\sum_{j=1}^{p} f(m_j/\widehat{\theta}_j) \Big/ \Big(-p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1})\Big) + 1}{p^{-1}\sum_{j=1}^{p} f(m_j/\mathbf{p}_{1j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1j}) \Big/ \Big(-p^{-1}\log\det(\boldsymbol{\Sigma}_p\boldsymbol{\Sigma}_{\text{Ideal},n_1}^{-1})\Big) + 1},$$

which is less than or equal to 1 almost surely if we can show that

$$p^{-1}\sum_{j=1}^{p} f(m_j/\widehat{\theta}_j) \xrightarrow{a.s.} 0. \tag{6.21}$$

To this end, for ξ_j lying between 1 and $m_j/\widehat{\theta}_j$ for each $j = 1, \ldots, p$, we have

$$\begin{aligned}
\left|p^{-1}\sum_{j=1}^{p} f(m_j/\widehat{\theta}_j)\right| &= \left|p^{-1}\sum_{j=1}^{p}\Big(f(1) + f'(\xi_j)(m_j/\widehat{\theta}_j - 1)\Big)\right| \leq p^{-1}\sum_{j=1}^{p}\frac{|\xi_j - 1|\cdot|m_j/\widehat{\theta}_j - 1|}{|\xi_j|} \\
&= p^{-1}\sum_{j=1}^{p}\frac{|\widehat{\theta}\xi_j - \widehat{\theta}_j|\cdot|m_j - \widehat{\theta}_j|}{|\widehat{\theta}_j|\cdot|\widehat{\theta}_j\xi_j|} \\
&\leq \frac{\max_{1\leq j\leq p}(\widehat{\theta}_j - m_j)^2}{\min_{1\leq j\leq p}(m_j - \max_{1\leq j\leq p}|\widehat{\theta}_j - m_j|)^2}.
\end{aligned} \tag{6.22}$$

We have $\min_{1\leq j\leq p} m_j \geq \lambda_{\min}(\boldsymbol{\Sigma}_p) > 0$, where $\lambda_{\min}(\boldsymbol{\Sigma}_p)$ is the smallest eigenvalue of $\boldsymbol{\Sigma}_p$ which is uniformly bounded away from 0 by Assumption (A2). Also, by (6.18), we have for each $j = 1, \ldots, p$,

$$P_{\mathbf{Y}_1}(|\widehat{\theta}_j - m_j| \geq k \max_{1\leq j\leq p}\beta_{n_2}^{(j)}) \leq 2p^{-3}.$$

Hence by the union sum inequality,

$$P_{\mathbf{Y}_1}(\max_{1\leq j\leq p}|\widehat{\theta}_j - m_j| \geq k \max_{1\leq j\leq p}\beta_{n_2}^{(j)}) \leq 2p^{-2}.$$

We can prove that $\max_{1\leq j\leq p}|\widehat{\theta}_j - m_j| \xrightarrow{a.s.} 0$ through the Borel-Cantelli's lemma since $\sum_{n_2\geq 1} 2p^{-2} < \infty$ under our settings, if we can show that $\max_{1\leq j\leq p}\beta_{n_2}^{(j)} = o(1)$. But using arguments similar to (6.19),

$$\max_{1\leq j\leq p}\beta_{n_2}^{(j)} = \begin{cases} O\left(\|\boldsymbol{\Sigma}_p\|\left(\frac{\log p}{n_2}\right)^{1/6}\right), & 1.2 < q_0 \leq 2; \\ O\left(\|\boldsymbol{\Sigma}_p\|\left(\frac{\log p}{n_2}\right)^{5(q_0-1)/6}\right), & 1 < q_0 \leq 1.2. \end{cases} \tag{6.23}$$

Hence by the assumptions on the two scenarios in the Theorem, we have $\max_{1\leq j\leq p}\beta_{n_2}^{(j)} = o(1)$, and hence $\max_{1\leq j\leq p}|\widehat{\theta}_j - m_j| \xrightarrow{a.s.} 0$, which shows also (6.21), that $p^{-1}\sum_{j=1}^{p} f(m_j/\widehat{\theta}_j) \xrightarrow{a.s.} 0$, through (6.22). The result $\max_{1\le j\le p}|\widehat{\theta}_j - m_j| \overset{a.s.}{\to} 0$ also shows that $\widehat{\mathbf{\Sigma}}_{n_1}$ is almost surely positive definite as $n\to\infty$ since m_j is uniformly bounded away from 0 by Assumption (A2). This completes the proof of the Theorem. $\square$

Lemma 4. *Assume the settings in Theorem 4. Define* $v_j := E_{\mathbf{Y}_1}^{1/q_0}|w_{ji} - m_j|^{q_0}$ *and* $\widehat{v}_j := E_{\mathbf{Y}_1}^{1/q_0}|(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2 - E_{\mathbf{Y}_1}(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2|^{q_0}$*, where* $w_{ji} := (\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu}))^2$ *and* $m_j := \mathbf{p}_{1j}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j}$*. Let* $\widehat{\alpha}$ *be defined as in (6.13). Then we have the following three results: For large enough* n*,*

I.

$$\exp\left(C_\psi\widehat{\alpha}\sum_{i=1}^{n_2}(2|\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| + |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|)|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|\right) \le p^C,\ j = 1,\ldots,p;$$

II.

$$\frac{\max_{1\le j\le p} v_j^{q_0}}{\max_{1\le j\le p} b_{nj}\widehat{v}_j^{q_0}} \le C';$$

III.

$$\max_{1\le j\le p}\widehat{v}_j^2 \le C''\max_{1\le j\le p}(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j})^2.$$

All three results hold almost surely, with C, C' *and* C'' *all positive constants independent of any indices.*

Proof of Lemma 4. We first apply Lemma 2 for $\widehat{\phi}_j$, $j = 1,\ldots,p$. Define $\widetilde{v}_{0j} := E_{\mathbf{Y}_1}^{1/2}(\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{21} - \boldsymbol{\mu}))^2$, $\widetilde{u}_j := \widetilde{b}_{nj}\widetilde{v}_{0j}$, $\widetilde{u} := \max_{1\le j\le p}\widetilde{u}_j$, where $\widetilde{b}_{nj} > 0$ is a constant unknown to us, and $\{\widetilde{b}_{nj}\}$ is a double array of constants uniformly bounded from 0 and infinity. Then we put $\delta = p^{-3}$, and define

$$\widetilde{\alpha} := \left(\frac{2}{(1+\lambda)}\right)^{1/2}\frac{1}{\widetilde{u}}\left(\frac{3\log p}{n_2}\right)^{1/2}.$$

Then $\widehat{\phi}_j$ is the solution to $\sum_{i=1}^{n_2}\psi(\widetilde{\alpha}(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \theta)) = 0$, where ψ can be ψ_2, $\varphi_{1,\lambda}$ or $\varphi_{2,\lambda}$. Following Theorem 3, noting that $q_0 = q = 2$ (we know the existence of second order moments), define

$$\begin{aligned}\widetilde{\beta}^{(j)}_{n_2,2,\widetilde{u}} &:= \widetilde{u}\left(\frac{(1+\lambda)}{2}\right)^{1/2}\left(\frac{3\log p}{n_2}\right)^{1/2}\left(\frac{\widetilde{v}_{0j}^2}{\widetilde{u}^2} + 1\right)\\ &\le \widetilde{u}\left(\frac{1+\lambda}{2}\right)^{1/2}\left(\frac{3\log p}{n_2}\right)^{1/2}\left(\frac{1}{b_{nj}^2} + 1\right) \le \widetilde{C}\left(\frac{\log p}{n_2}\right)^{1/2}\end{aligned}$$

for some constant $\widetilde{C} > 0$. Then for $1 < k \le 2$, with

$$\frac{3\log p}{n_2} \le \frac{2(k-1)}{(1+\lambda)k^2}\left(1 + \frac{1}{\min_j b_{nj}^2}\right)^{-1} \le \frac{2(k-1)}{(1+\lambda)k^2}\left(1 + \frac{\widetilde{v}_{0j}^2}{\widetilde{u}^2}\right)^{-1},$$

Lemma 2 gives, for each $j = 1, \ldots, p$,

$$P_{\mathbf{Y}_1}(|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}| \geq k\widetilde{\beta}_{n_2,2,\widetilde{u}}^{(j)}) \leq 2p^{-3},$$

so that for each $j = 1, \ldots, p$,

$$P_{\mathbf{Y}_1}(|\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}| \geq k \max_{1\leq j\leq p} \widetilde{\beta}_{n_2,2,\widetilde{u}}^{(j)}) \leq 2p^{-3}.$$

The union sum inequality then gives

$$P_{\mathbf{Y}_1}(\max_{1\leq j\leq p} |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}| \geq k \max_{1\leq j\leq p} \widetilde{\beta}_{n_2,2,\widetilde{u}}^{(j)}) \leq 2p^{-2}. \tag{6.24}$$

Under $p/n \to c > 0$, with assumption $\log p/n_2 \to 0$, we have $\max_{1\leq j\leq p} \widetilde{\beta}_{n_2,2,\widetilde{u}}^{(j)} \to 0$ as $n \to \infty$. And since n_2 has the same order as n, we have

$$\sum_{n_2\geq 1} 2p^{-2} = O\left(\sum_{n_2\geq 1} n_2^{-2}\right) = O(1).$$

Hence by the Borel-Cantelli's lemma, we have that given $\mathbf{Y}_1$,

$$\widehat{\phi}_j \overset{a.s.}{\to} \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu} \text{ uniformly for all } j, \quad \text{with a rate } \left(\frac{\log p}{n_2}\right)^{1/2}.$$

Hence on an almost sure set $\mathcal{M}$, for some constant $B > 0$, we have for large enough n and given $\mathbf{Y}_1$,

$$\max_{1\leq j\leq p} |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}| \leq B\left(\frac{\log p}{n_2}\right)^{1/2}. \tag{6.25}$$

We prove result I first. Define

$$\begin{aligned} A_1 &:= 2C_\psi\widehat{\alpha}\sum_{i=1}^{n_2} |\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| \cdot |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}|, \\ A_2 &:= C_\psi\widehat{\alpha}n_2 \max_{1\leq j\leq p} (\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu})^2. \end{aligned}$$

Using (6.25), on $\mathcal{M}$ and for large enough n, we have

$$A_2 \leq C_\psi B^2 n_2 \left(\frac{6}{1+\lambda}\right)^{1/2} \frac{1}{\max_j b_{nj}\widehat{v}_j} \left(\frac{\log p}{n_2}\right)^{11/6} = O((\log p)^{11/6} n_2^{-5/6}) = o(1),$$

since $\{b_{nj}\}$ is uniformly bounded from 0, and $\widehat{v}_j$ as a measure of spread of $\{(\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{y}_{2i} - \widehat{\phi}_j)^2\}_{i=1,\ldots,n_2}$ is also uniformly bounded away from 0 since $\mathbf{\Sigma}_p$ has eigenvalues uniformly bounded away from 0 by Assumption (A2). The above proves that $A_2 \overset{a.s.}{\to} 0$.

For A_1, for a constant $b > 0$,

$$\begin{aligned}
P_{\mathbf{Y}_1}(A_1 \geq C\log p) &\leq P_{\mathbf{Y}_1}\left(b\left(\frac{\log p}{n_2}\right)^{1/2+1/3} |\widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}| \sum_{i=1}^{n_2} |\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})| \geq C\log p\right) \\
&\leq I_1 + I_2, \quad \text{where} \\
I_1 &:= P_{\mathbf{Y}_1}(\max_{1\leq j\leq p} |\widehat{\phi}_j - \mathbf{p}_{1j}^T\boldsymbol{\mu}| \geq k\widetilde{C}(\log p/n_2)^{1/2}), \\
I_2 &:= P_{\mathbf{Y}_1}(b(\log p/n_2)^{5/6} \sum_{i=1}^{n_2} f_{ji} \geq C\log p \cdot k^{-1}\widetilde{C}^{-1}(\log p/n_2)^{-1/2}).
\end{aligned}$$

where $f_{ji} := |\mathbf{p}_{1j}^{\mathrm{T}}(\mathbf{y}_{2i} - \boldsymbol{\mu})|$. From (6.24), we have $I_1 \leq 2p^{-2}$. Consider $I_2 \leq I_{21} + I_{22}$, where

$$\begin{aligned}
I_{21} &:= P_{\mathbf{Y}_1}(b(\log p/n_2)^{5/6} \sum_{i=1}^{n_2} E_{\mathbf{Y}_1} f_{ji} \geq C\log p \cdot k^{-1}\widetilde{C}^{-1}(\log p/n_2)^{-1/2}/2), \\
I_{22} &:= P_{\mathbf{Y}_1}(b(\log p/n_2)^{5/6} \sum_{i=1}^{n_2} (f_{ji} - E_{\mathbf{Y}_1} f_{ji}) \geq C\log p \cdot k^{-1}\widetilde{C}^{-1}(\log p/n_2)^{-1/2}/2).
\end{aligned}$$

For I_{21}, since $E_{\mathbf{Y}_1} f_{ji} \leq (\mathbf{p}_{1j}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j})^{1/2} \leq \left\|\mathbf{\Sigma}_p\right\|^{1/2}$,

$$I_{21} \leq P(b(\log p/n_2)^{1/3} \max_{1\leq j\leq p} E_{\mathbf{Y}_1} f_{ji} \geq C/(2k\widetilde{C})) \leq P(\left\|\mathbf{\Sigma}\right\|^{1/2}(\log p/n_2)^{1/3} \geq C/(2kb\widetilde{C})) = 0$$

for large enough n, since $\left\|\mathbf{\Sigma}_p\right\| = O(p^\gamma)$ and $p^{3\gamma/2}\log p/n_2 = o(1)$ by assumptions in both scenario *i* or *ii*. For I_{22}, using Markov's inequality and that $p/n \to c > 0$ and $n_2/n \to 1/2$,

$$\begin{aligned}
I_{22} &= P_{\mathbf{Y}_1}\left(\frac{(\log p)^{1/3}}{n_2^{4/3}} \sum_{i=1}^{n_2} (f_{ji} - E_{\mathbf{Y}_1} f_{ji}) \geq \frac{C}{2bk\widetilde{C}}\right) \leq \frac{4b^2k^2\widetilde{C}^2(\log p)^{2/3}}{C^2 n_2^{8/3}} \cdot n_2 m_j \\
&= O(p^\gamma(\log p)^{2/3}n_2^{-5/3}) = O((\log n)^{2/3}n^{-(5/3-\gamma)}).
\end{aligned}$$

But $\sum_{n\geq 1}(\log n)^{2/3}n^{-(5/3-\gamma)} < \infty$ since $\gamma < 2/3$ in both scenarios, showing that $\sum_{n\geq 1} P(A_1 \geq C\log p) \leq \sum_{n\geq 1}(I_1 + I_{21} + I_{22}) < \infty$, and hence $A_1 \leq C\log p$ almost surely for large enough n. Replacing C in the proof by $C - \tau$ for a small $\tau > 0$, and with $A_2 \overset{a.s.}{\to} 0$ as well, result I is proved.

For result II and III, suppose 4th order moments were to exist for the data. Then by Jensen's inequality, with $\mathbf{A} := \boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j}\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p^{1/2}$,

$$
\begin{aligned}
v_j^2 &\le E_{\mathbf{Y}_1}(w_{ji} - m_j)^2 = E_{\mathbf{Y}_1}(\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p^{1/2}\mathbf{z}_i\mathbf{z}_i^{\mathrm{T}}\boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j})^2 - m_j^2 \\
&= \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p^{1/2}E(\mathbf{z}_i\mathbf{z}_i^{\mathrm{T}}\mathbf{A}\mathbf{z}_i\mathbf{z}_i^{\mathrm{T}})\boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j} - m_j^2 \\
&= m_j^2 + (\nu_4 - 3)\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p^{1/2}\mathrm{diag}(\mathbf{A})\boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j} + 2m_j^2 - m_j^2 \\
&\le (|\nu_4 - 3| + 2)m_j^2,
\end{aligned}
\tag{6.26}
$$

where $\nu_4 = E|z_{ij}^4|$. The third line used Lemma A.2 in the supplement of Li et al. (2019). The fourth line used the fact that, if we define $\mathbf{a} := (a_1, a_2, \ldots, a_p)^{\mathrm{T}} := \boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j}$, then

$$
\mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathrm{diag}(\mathbf{A})\boldsymbol{\Sigma}_p^{1/2}\mathbf{p}_{1j} = \mathbf{a}^{\mathrm{T}}\mathrm{diag}(\mathbf{a}\mathbf{a}^{\mathrm{T}})\mathbf{a} = \sum_{j=1}^p a_j^4 \le \left(\sum_{j=1}^p a_j^2\right)^2 = (\mathbf{a}^{\mathrm{T}}\mathbf{a})^2 = m_j^2.
$$

The above shows that

$$
\max_{1\le j\le p} v_j^2 \le c'' \max_{1\le j\le p} m_j^2 \tag{6.27}
$$

for some constant $c'' > 0$. Writing $\widehat{d}_j := \widehat{\phi}_j - \mathbf{p}_{1j}^{\mathrm{T}}\boldsymbol{\mu}$, we have for large enough n,

$$
\begin{aligned}
\widehat{v}_j^{q_0} &= E_{\mathbf{Y}_1}|(w_{ji} - m_j) - 2w_{ji}\widehat{d}_j + 2E_{\mathbf{Y}_1}(w_{ji}\widehat{d}_j) + \widehat{d}_j^2 - E_{\mathbf{Y}_1}\widehat{d}_j^2|^{q_0} \\
&\le 2^{q_0-1}\left(v_j^{q_0} + \max_{1\le j\le p} E_{\mathbf{Y}_1}|2w_{ji}\widehat{d}_j - 2E_{\mathbf{Y}_1}(w_{ji}\widehat{d}_j) - \widehat{d}_j^2 + E_{\mathbf{Y}_1}\widehat{d}_j^2|^{q_0}\right) \\
&\le 2^{q_0-1}(v_j^{q_0} + O(\max_{1\le j\le p} m_j^{q_0} \cdot (\log p/n_2)^{q_0/2})) \\
&= 2^{q_0-1}v_j^{q_0} + O((p^{2\gamma}\log p/n_2)^{q_0/2}).
\end{aligned}
$$

where the second line used the C_r inequality and the third line used (6.25) and similar arguments to (6.26). The above implies that for large enough n,

$$
\max_{1\le j\le p} \widehat{v}_j/v_j = O(1). \tag{6.28}
$$

But since both $\widehat{v}_j$ and v_j are uniformly bounded away from 0, the above also implies that

$$
\min_{1\le j\le p} \widehat{v}_j/v_j
$$

is uniformly bounded away from 0, so that

$$\frac{\max_{1\le j\le p} v_j^{q_0}}{\max_{1\le j\le p} b_{nj}\widehat{v}_j^{q_0}} \le \max_{1\le j\le p} \frac{v_j^{q_0}}{b_{nj}\widehat{v}_j^{q_0}} \le \frac{1}{\min_j b_{nj}\cdot \min_j(\widehat{v}_j/v_j)^{q_0}} = O(1),$$

which is result II. Result III follows from (6.27) and (6.28):

$$\max_{1\le j\le p} \widehat{v}_j^2 = \max_{q\le j\le p} v_j^2 \cdot \widehat{v}_j^2/v_j^2 \le \max_{1\le j\le p} {v_j}^2 \cdot \max_{1\le j\le p} \widehat{v}_j^2/v_j^2 = O(\max_{1\le j\le p} v_j^2) \le C'' \max_{1\le j\le p} m_j^2.$$

This completes the proof of the lemma. □

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